Proxmox Storage DRS -- Operator Manual

Installing, configuring and running it safely

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What you need

- **Proxmox VE 9.2**, or a management host with network access to a PVE 9.x cluster's API (tcp/8006) — the tool does not need to run on a cluster member.
- **A Prometheus-compatible backend** already receiving PVE's per-disk `blockstat` metrics via the InfluxDB external metric server — plain Prometheus, VictoriaMetrics, or gigapipe on ClickHouse, which is what this project is dogfooded against. If your cluster's dashboards already show per-VM disk I/O, the export side is already on. That is not the same as all six counters having survived the trip, which is a real and silent failure mode: read `05-metrics-pipeline.md` before writing the `metrics` block, and run `pve-storage-drs verify-metrics` before relying on any plan. No new exporter or collector is required.
- **A PVE user or API token** with read access to VM and storage inventory and permission to call `move_disk`. An API token is preferred for unattended operation (`execution.mode: auto`); a username/password pair works for interactive use. See "Setting up the PVE credential" below for the exact privileges — the obvious-looking "just grant Audit everywhere" is not enough, and the gap is silent rather than an error.

Setting up the PVE credential

Grant, on **every storage the tool will manage** (the shared, image-holding storages in your `groups`, not backup targets):

- `Datastore.Audit` **and** **`Datastore.Allocate`** — not Audit alone. `GET /nodes/{node}/storage/{storage}/content` which the tool uses to cross-check each disk's real allocated size, silently returns an empty list under Audit-only permission: a clean `200` response with no content and no error, indistinguishable from "this storage genuinely has nothing on it" unless you already know to be suspicious. Audit alone is enough for every *other* read call the tool makes; this one endpoint is the exception, verified against a live PVE 9.2.11 cluster (see `IMPLEMENTATION_PLAN.md` section 3.5's note). Getting this wrong does not make the tool fail loudly — it makes it under-count what already occupies each storage, which quietly erodes the **snapshot reserve**: the free capacity `pve-storage-drs` always keeps clear on every managed storage (sized from `snapshot_reserve.factor`/`free_space.soft` — see `10-configuration.md`) so that a snapshot, or a disk mid-migration existing on two storages at once, never runs a LUN out of space. An under-counted storage looks to have more headroom than it really does, so the tool plans and executes migrations that erode a safety margin it believes it is still protecting.
- Grant it **at each storage's own path** (`/storage/<id>`), not only at the parent `/storage` — that is the grant confirmed to work.
- Also add `VM.Audit` cluster-wide (`/`) for VM inventory and config, which `PVEAuditor` already includes if you use that built-in role as a base.

If you are using an API token (recommended — see above), Proxmox's privilege separation means **the token has its own, separate ACL entries from its owning user**. A role granted to the user alone has no effect on the token, and a role granted to the token alone has no effect either unless the user has it too: the token's effective permission is the *intersection* of the two. Grant the same roles to both the user and `user@realm!tokenid`, or disable privilege separation on the token (Datacenter → Permissions → API Tokens → uncheck "Privilege Separation") so it always matches its user's permissions exactly.

A concrete, minimal setup: create a role (e.g. `pve-storage-drs`) with `Datastore.Allocate,Datastore.Audit,VM.Audit`, then add it as an ACL entry for both the user and the token at `/` for `VM.Audit`, and at each managed storage's `/storage/<id>` path for the datastore privileges.

Two further, VM-level privileges are needed once `apply` executes a move — `VM.Config.Disk` and `VM.Migrate`, both cluster-wide (`/`) or at least on every VM whose disks live in a managed group, granted

to both the user and the token exactly as above. Not needed for `verify-metrics`, `show-load` or `verify-storages`, which only read; `apply` needs them for `move_disk` itself, in every `execution.mode` (`30-safety-and-status.md`). Grant them now alongside the privileges above so the credential does not need revisiting later; if you set up the credential read-only for now, add `VM.Config.Disk,VM.Migrate` to the role before you first run `apply`.

Installing the package

On a Debian trixie host (which is what Proxmox VE 9.x is built on). Every release is published as a ready-built package on the project's GitHub releases page, https://github.com/bzed/proxmox-storage-drs/releases: one architecture-independent `pve-storage-drs_<version>_all.deb`, with a `SHA256SUMS` file beside it. There is no apt repository yet, so download the `.deb` and install it with `apt`, which fetches the dependencies from the trixie archive the host already uses:

```
VERSION=0.1.21      # the newest release on the releases page
BASE=https://github.com/bzed/proxmox-storage-drs/releases/download/debian/$VERSION
wget "$BASE/pve-storage-drs_${VERSION}_all.deb" "$BASE/SHA256SUMS"
sha256sum -c SHA256SUMS
apt install ./pve-storage-drs_${VERSION}_all.deb
```

The package is not signed. `SHA256SUMS` comes from the same page as the `.deb`, so it detects a corrupted download, not a tampered release. A release is published only after the project's CI has built the package in a clean Debian trixie container, checked it with lintian, installed it and run its tests there. If you keep a local apt repository, put the `.deb` into it and `apt install pve-storage-drs` and `apt upgrade` work as usual; from a source checkout, `make deb` builds the same package.

This installs the `pve-storage-drs` executable, its manpage, the example configuration at `/usr/share/doc/pve-storage-drs/examples/drs.example.yaml`, and the whole documentation set under `/usr/share/doc/pve-storage-drs/`: this manual as `manual/*.md`, the internals guide as `internals/*.md` and the specification as `IMPLEMENTATION_PLAN.md`, each also as a typeset PDF. The Markdown is deliberately left uncompressed — a cluster node has no GUI and often no way to get a PDF off it, so `less /usr/share/doc/pve-storage-drs/manual/20-verifying-metrics.md` has to work on the machine that has the problem.

`coinor-cbc` and `python3-pulp` are `Depends`: `apt install pve-storage-drs` always gets a real MILP solver, no separate step needed. The tool still plans without one — using the dependency-free heuristic of `IMPLEMENTATION_PLAN.md` section 5.5 — but that is no longer what a Debian install lands on by default; it is what a non-Debian `pip install` with no solver extra, or a solve that fails or times out, falls back to.

Where the configuration lives

`/etc/pve/drs.yaml`. On a cluster member this path is on `pmxcfs`, the cluster filesystem, so the file is automatically the same on every node — there is exactly one copy and no question of which one is authoritative. A management host that is not itself a cluster member simply keeps an ordinary file at that same path.

Two things follow from that placement, and both matter before you put a password in the file:

- **An edit is live on every node the instant it is saved.** There is no staging step. This is why configuration mistakes are always hard failures (see `30-safety-and-status.md`) rather than warnings, and why dry-run is the default execution mode.
- **Files under `/etc/pve` are group-readable by `www-data`** — the PVE web server's user. Check `ls -l /etc/pve/drs.yaml` on your cluster before deciding whether a plaintext `password:` is acceptable; the safer choice is an API token with `PVE_TOKEN_SECRET` set in the environment (or a systemd credential) rather than written into the file at all.

Copy `/usr/share/doc/pve-storage-drs/examples/drs.example.yaml` to `/etc/pve/drs.yaml` and edit it — every key is commented with its default and what it does. The full reference is `10-configuration.md`.

Where the configuration is *not*

`state.path` (default `/var/lib/pve-storage-drs/state.json`) is deliberately **not** on `/etc/pve`. It is rewritten on every run, `pmxcfs` writes need cluster quorum, and the state it holds — cooldown timestamps, the load vector at the last balance, the run lock — is meaningful only to the one host that is actually running the timer. See `10-configuration.md#statepath`.

Running the timer on exactly one host

Putting the configuration on `pmxcfs` does not make the tool cluster-aware. `state.json` is node-local: each node would keep its own cooldowns, its own drift baseline and its own run lock, which do not coordinate with each other. **Enable the systemd timer on exactly one host.** The one thing that *does* cross the cluster is a startup scan for in-flight `move_disk` UPIDs owned by the DRS user, which is what keeps two accidental instances from actively conflicting (they degrade to "slow and redundant" instead).

First steps after installing

1. Read `05-metrics-pipeline.md` — where the six per-disk counters come from, and how a Telegraf-to-Prometheus transport can drop one of them without ever logging an error.
2. Read `10-configuration.md` and adjust the storage groups, thresholds and weights for your cluster.
3. `pve-storage-drs -c /etc/pve/drs.yaml verify-metrics` — confirms the configured metric and label names actually exist in your backend, and cross-checks the six metrics against each other. See `20-verifying-metrics.md`.
4. See `30-safety-and-status.md` for exactly which commands this build supports end to end today.

Where the numbers come from, and how a transport loses them

`pve-storage-drs` collects nothing itself. It ships no exporter, installs no agent on your nodes and opens no connection to a VM. Every number in every plan comes out of a Prometheus-compatible backend you already run, via a pipeline PVE already has.

Read this page before you write the `metrics` block in `10-configuration.md`. The pipeline is where this tool's inputs get quietly damaged, and a damaged input produces a confident plan rather than an error.

The six counters

Per VM disk, per push:

```
rd_operations   wr_operations         how many I/Os
rd_bytes        wr_bytes              how much data
rd_total_time_ns  wr_total_time_ns    how long the device was busy
```

They are QEMU's own `query-blockstats` figures. `pvestatd` already gathers them on every status push — it calls `vmstatus(undef, 1)`, and the `full` flag is what makes it issue the QMP query — and PVE's **InfluxDB external metric server** already exports the whole stats hash without filtering.

`PVE/Status/InfluxDB.pm` flattens that nested hash so the first level becomes the measurement (`blockstat`), the second becomes a **tag** (`instance=scsi0`), and the leaves become fields. A series therefore arrives looking roughly like:

```
blockstat_rd_operations{vmid="101", instance="scsi0", nodename="pve01", host="db-01"}
```

The device id being a **label** rather than part of the metric name is the whole reason this approach works: it is what makes per-disk `rate()` and aggregation across drives possible at all.

Note the label names in that example. PVE's own tag for the drive id is `instance`, which **collides with Prometheus's own scrape-target label of the same name**, and many Telegraf configurations rename or overwrite one of the two. That is why the tool never hardcodes them: `metrics.labels.vmid`, `.device` and `.node` are configuration keys naming whatever labels your deployment actually produces, defaulting to `vmid`, `instance` and `nodename`. `verify-metrics` warns when `.device` is still `instance` — not because that is wrong, but because it is the one name where you have to confirm which of the two survived before anything joins on it.

So the data exists cluster-wide before this tool is installed, and there is nothing new to deploy. `IMPLEMENTATION_PLAN.md` §3.1 carries the source excerpts if you want to confirm any of the above against `pve-manager` and `qemu-server` yourself.

Two things deliberately not used

- **RRD.** `GET /nodes/{node}/qemu/{vmid}/rrddata` exposes only VM-aggregate `diskread`/`diskwrite` in bytes per second — no per-disk breakdown, no operation counts, no latency. It cannot answer the question this tool asks.
- **The OpenTelemetry metric server** (new in PVE 9.1), despite being the more modern-looking option. It appends a nested key to the metric *name* rather than carrying it as an attribute, so `blockstat.scsi0.rd_operations` becomes `proxmox_vm_blockstat_scsi0_rd_operations_total`. The device is baked into the name, which makes aggregation across devices impossible without `label_replace` gymnastics and makes metric-name cardinality unbounded. The InfluxDB path is strictly better here.

The failure mode: a transport that cannot carry strings

This is the most common way to get a wrong answer out of `pve-storage-drs`, and it never announces itself.

PVE's InfluxDB output speaks **InfluxDB line protocol**, which carries **string** fields alongside numeric ones. Two properties of that protocol combine badly with a Prometheus-shaped destination:

1. **A field's type is fixed by its first write.** Once a given series' field has arrived as a string, it is a string for that series from then on — permanently, not until the next push corrects it.
2. **Prometheus and OpenMetrics have no string sample type.** A sink that cannot represent a string field does the only thing available to it: it drops the field.

Put together: one non-numeric value, once, possibly months ago, for one disk, is enough to make one of the six counters vanish for that series while the other five keep working perfectly. Nothing logs an error. Your dashboards keep drawing, because they rarely plot all six.

What `pve-storage-drs` then sees is a disk that does no writes, or a disk whose device was never busy. The load model has no way to tell that apart from a genuinely idle disk — the data does not say "missing", it says nothing at all — so it plans around a number that is simply false.

Telegraf with a Prometheus remote-write output

This is the prominent example, and it is prominent because it is what most people build first:

```
inputs.influxdb_listener  →  (no processors)  →  outputs.prometheus_remote_write
```

`outputs.prometheus_client` and any other OpenMetrics-shaped output behave the same way, as does any other bridge from PVE's InfluxDB output into a Prometheus data model. **Unless the pipeline is configured deliberately** — converting the string fields, or dropping them explicitly with a processor so you know what you lost — it will silently discard fields, and you will find out from a plan that does not make sense rather than from a log line.

This is not an argument against Telegraf. It is an argument for configuring its processors on purpose, and for running `verify-metrics` afterwards rather than assuming.

What catches it

`verify-metrics` cross-checks the full `(vmid, device)` set that each of the six metrics reports against the other five, and warns — naming the metric and the disk — whenever one metric's set is a strict subset of another's. This costs your Prometheus nothing extra: it reuses the six instant queries that step of `verify-metrics` was already making in order to print a sample series.

Treat that warning as a real defect in the pipeline, not as noise. Fixing the pipeline does not retroactively restore the samples you never received, and by point 1 above it does not necessarily restore future ones either: as long as the affected series keeps arriving under the same identity with the field pinned to a string type, it stays dropped. Re-check with `verify-metrics` after any change rather than assuming.

A reference implementation that is well tested

gigapipe with ClickHouse as the database, fed directly from PVE's InfluxDB output.

gigapipe accepts **the InfluxDB protocol exactly as PVE exports it** on the ingest side, and serves the Prometheus HTTP query API — `/api/v1/query`, `/api/v1/query_range`, `/api/v1/label/<name>/values`, which is the entire surface this tool uses — on the read side. So PVE's own metric server writes to it directly and `pve-storage-drs` reads from it directly: there is no Telegraf, no output plugin and no format bridge anywhere in the path, and so nothing that can drop a field on type grounds.

It is the backend this project is dogfooded against. Both bundles in `tests/corpus/` were captured from a cluster running it and record it as their `prometheus_backend`, so the regression suite runs against real gigapipe-shaped data on every `make check`.

To use it, point `prometheus.url` at gigapipe's query endpoint. Nothing else in this manual changes — including at the `metrics.step`/`metrics.rate_window` default: a recently-updated gigapipe was found, live, to return zero series for a range query whose step reaches its `rate()` window's duration (exactly that default), and this tool works around it unconditionally on every backend rather than asking you to change your configuration for gigapipe's sake. See `metrics.step` in `docs/manual/10-configuration.md` for what the workaround changes about what actually gets fetched.

gigapipe also ingests OpenTelemetry, but **that path is untested here** and is not the way to use it with this tool. Feeding it from PVE's OpenTelemetry metric server instead of the InfluxDB one would reintroduce the device-in-the-metric-name problem described above, whatever the backend does with it afterwards.

Other backends

Plain **Prometheus** and **VictoriaMetrics** both work. Every query this tool issues — `rate()`, `quantile_over_time`, subquery ranges, and `/api/v1/label/__name__/values` — is supported by all three backends, and `prometheus.url` is the only setting that distinguishes them.

Whatever you use, the pipeline in front of it is still yours to get right, and `verify-metrics` is still the thing that tells you whether you did.

What to do next

1. Work out which transport your cluster actually uses today. The PVE side is the external metric server — `/etc/pve/status.cfg`, which is what Datacenter → Metric Server writes; if there is no InfluxDB entry there, nothing described on this page is happening yet and that is the first thing to set up.

   Do **not** take an existing "VM disk I/O" dashboard as evidence that it is. Such a dashboard may be drawing PVE's RRD data, or series from `prometheus-pve-exporter`, neither of which carries

per-disk `blockstat` at all. Query your backend for `blockstat_rd_operations` directly and look at whether the result carries a per-drive label.

2. Set the `metrics` block in `/etc/pve/drs.yaml` to match the names that transport produces — they are configuration, not constants, because Telegraf's measurement/field join character and tag renaming vary per deployment. `10-configuration.md` documents every key with its default.
3. Run `verify-metrics` and read every finding.

Configuration reference

Every key `pve-storage-drs` reads is documented here: type, unit, default, what it interacts with, and what happens if you set it too high or too low. A key that exists in the schema but not here, or here but not in the schema, is a bug — `tests/unit/test_documentation.py` asserts the two never drift apart.

`config/drs.example.yaml` in the source tree (installed to `/usr/share/doc/pve-storage-drs/examples/drs.example.yaml`) carries the same defaults with inline comments; this page is the full explanation each comment points back to.

`schema_version`

Integer, required, no default.

The config format's version. `pve-storage-drs` refuses to start on a `schema_version` whose **major** number it does not understand, naming the version it does understand in the error, rather than guessing at fields from a future incompatible format. Currently `1`.

`proxmox` — the cluster API connection

`proxmox.host`

String, required, no default.

Any cluster member's hostname or IP. The API is reached at `https://<host>:<port>/api2/json`.

`proxmox.port`

Integer, default `8006`.

The PVE API port. Change this only if you have put something unusual (a reverse proxy) in front of the API.

`proxmox.verify_ssl`

Boolean, default `true`.

Whether to verify the API's TLS certificate. Set `false` only for a self-signed certificate you cannot otherwise trust, and prefer `proxmox.ca_file` instead where possible — disabling verification also disables protection against a compromised network path to every node's API.

`proxmox.ca_file`

String path or `null`, default `null`.

A CA bundle to verify the API's certificate against, when it is not signed by a certificate already trusted by the host running `pve-storage-drs`.

`proxmox.auth.username`

String or `null`.

A PVE user, e.g. `drs@pve`. Required unless `proxmox.auth.token_id` is set. Username/password authentication is what the original requirement asked for; an API token (below) is preferred for `execution.mode`:

`auto` because it needs no interactive ticket refresh and can be scoped tightly: the tool's actual minimum privilege is `Datastore.Audit` **and** `Datastore.Allocate` on every storage in the config's `groups`, plus `VM.Config.Disk` and `VM.Migrate` (or an equivalent custom role) on the VMs it may move. `Datastore.Audit` alone is not enough — PVE's per-volume content listing silently returns an *empty* result with no error when only Audit is granted, which would make every tracked disk look untracked rather than raising anything you'd notice. Grant the role per storage rather than relying on propagation from the parent `/storage` path, and remember that an API token's effective permission is the **intersection** of the user's own grants and the token's own: both need the grant, or neither has it. (`IMPLEMENTATION_PLAN.md` section 3.5.)

`proxmox.auth.password`

String or `null`, default `null`, environment: `PVE_PASSWORD`.

Leave this `null` in the file and set `PVE_PASSWORD` in the environment (or a systemd credential) instead, especially when the config lives on `/etc/pve` and is therefore group-readable by `www-data`. A password given in the file is used as-is and is **not** overridden by the environment variable — see `00-installation.md` for why the file's own permissions matter here.

`proxmox.auth.token_id`

String or `null`, default `null`.

An API token id, e.g. `drs@pve!balancer`. Set this instead of `username`/`password` for unattended operation; a token needs no ticket refresh loop and can be revoked independently of the user's password.

`proxmox.auth.token_secret`

String or `null`, default `null`, environment: `PVE_TOKEN_SECRET`.

The secret half of the API token, with the same environment-variable preference as `proxmox.auth.password`.

`proxmox.ticket_refresh_seconds`

Duration, default `3000` (50 minutes).

How often a username/password ticket is refreshed. PVE tickets last approximately two hours; refreshing well before that avoids a mid-run authentication failure. Not used for API-token authentication, which needs no ticket. Implemented by overriding `proxmoxer`'s own (otherwise fixed-at-3600s) refresh interval after login, since `proxmoxer` has no constructor option for it; if a ticket somehow still gets rejected mid-run regardless (a long `confirm`-mode wait, a suspended process, a clock jump — see `50-pve-api.md`), the client transparently logs in again from scratch and retries the one call that failed, once, before giving up.

`proxmox.read_workers`

Integer, default `12`.

Size of the bounded thread pool used to fetch per-VM configuration — `GET /nodes/{node}/qemu/{vmid}/config` has no batch form, so this is what keeps a several-hundred-VM cluster's topology read from being serial. Raising it trades API server load for wall-clock time; 8–16 is the range the plan suggests. Too high a value on a small PVE API server can itself become the bottleneck. Only the per-VM fetch is parallelized; the cluster-wide and per-storage calls (`50-pve-api.md`) still run once each, sequentially, before it starts.

`prometheus` — the metrics source

`prometheus.url`

String, required, no default.

The base URL of your Prometheus, e.g. `http://prometheus.example.com:9090`.

`prometheus.timeout_seconds`

Duration, default `30`.

Per-request timeout against Prometheus. Too low a value on a busy or long-range query makes `verify-metrics` and the load model fail spuriously; too high a value delays noticing that Prometheus is actually unreachable.

`prometheus.username` / `prometheus.password`

String or `null`, default `null`.

HTTP basic auth credentials, if your Prometheus requires them. Ignored when `prometheus.bearer_token` is set.

`prometheus.bearer_token`

String or `null`, default `null`.

A bearer token for Prometheus instances behind an auth proxy that expects one. Takes precedence over basic auth when both are set.

`metrics` — Telegraf/InfluxDB name mapping

Telegraf's naming is deployment-specific (`IMPLEMENTATION_PLAN.md` section 3.3), so every name below is configuration, never a constant. Run `pve-storage-drs verify-metrics` after changing any of them — see `20-verifying-metrics.md`.

`metrics.read_ops`

String, default `blockstat_rd_operations`.

The Prometheus metric name for per-disk read operations/second (raw counter, before `rate()`).

`metrics.write_ops`

String, default `blockstat_wr_operations`.

As `metrics.read_ops`, for writes.

`metrics.read_bytes`

String, default `blockstat_rd_bytes`.

The Prometheus metric name for per-disk bytes read (raw counter).

`metrics.write_bytes`

String, default `blockstat_wr_bytes`.

As `metrics.read_bytes`, for writes.

`metrics.read_time_ns`

String, default `blockstat_rd_total_time_ns`.

The Prometheus metric name for cumulative read I/O time in nanoseconds. This is the **primary** load signal by default (`load_weights.iotime`): by Little's law, `rate(rd_total_time_ns + wr_total_time_ns) / 1e9` is the average number of I/O requests in flight on that disk — dimensionless, additive across disks on the same storage, and directly comparable between storages of different size and speed. It also *self-weights* reads against writes automatically: an operation that takes eight times as long to service (a large sequential write against a small random read, say) counts eight times as much, with no manual read/write tuning needed. Pure IOPS treats every operation as equal and so under-counts large sequential load; pure throughput does the reverse and under-counts small random load. `ops` and `bytes` (see `load_weights` below) remain available as additional weighted terms for a policy the array's own timings do not capture. (`IMPLEMENTATION_PLAN.md` section 4.)

`metrics.write_time_ns`

String, default `blockstat_wr_total_time_ns`.

As `metrics.read_time_ns`, for writes.

`metrics.labels.vmid`

String, default `vmid`.

The Prometheus label carrying the VM id.

`metrics.labels.device`

String, default `instance`.

The Prometheus label carrying the drive id (`scsi0`, `virtio0`, ...). **Confirm this with `verify-metrics`.** PVE's own tag of this name collides with Prometheus's own scrape-target `instance` label, and many Telegraf configurations rename or overwrite it; `verify-metrics` warns loudly when this is still set to the literal string `instance`.

`metrics.labels.node`

String, default `nodename`.

The Prometheus label carrying the PVE node name — what the auto-derived filter below is built from: the tool fetches this cluster's own node list from the PVE API and scopes every query to `<this label>=~"<node1>|<node2>|..."`, so a Prometheus shared by more than one PVE cluster (or by anything else emitting a same-named metric) cannot silently sum in a same-numbered vmid from somewhere else. This is the default filter for every run; `verify-metrics` never applies it — see `metrics.extra_selector` below.

Confirm what's actually there with `verify-metrics` before assuming the default is right for your Prometheus: it reports each configured metric's sample series labels so you can check this key names a real one.

`metrics.extra_selector`

String or `null`, default `null` (auto).

Overrides the automatic node-scoping filter above with a raw PromQL label matcher, inserted verbatim into every query's vector selector (e.g. `{metrics.extra_selector}` becomes part of `rate(blockstat_rd_operations{cluster="mycluster"}[5m])`). Set this when the node-list filter isn't quite what you need — your own Telegraf/InfluxDB tagging scheme doesn't put the exact PVE node name in `metrics.labels.node`, you want to scope by something else (a cluster-naming tag your deployment happens to carry, say), or the restriction needed is something else entirely. Write exactly what your label scheme needs, e.g. `nodename=~"pve01|pve02|pve03"` or `cluster="mycluster"`; the value is used as-is, with no further quoting or escaping. `pve-storage-drs verify-metrics` applies this override too when set, but never derives its own filter automatically — it is deliberately independent of the PVE API entirely.

There is deliberately no "cluster label" auto-scoping tier above the node-list default: an earlier revision of this tool assumed such a label was standard practice across this project's deployments, which turned out to be wrong — nothing in the Telegraf/InfluxDB pipeline guarantees one. Use `metrics.extra_selector` for that instead, if your Prometheus actually carries such a label.

`metrics.rate_window`

Duration, default `5m`.

The inner range passed to `rate()`. Must be at least four times `metrics.pvestatd_push_interval`, or `rate()` sees too few points to be meaningful — `verify-metrics` measures the *observed* sample spacing of

a live series and errors if this rule is violated against reality, not just against the declared pvestatd_push_interval.

metrics.step

Duration, default 5m.

The sampling step used for range queries (forecasting, coverage checks). Independent of metrics.rate_window, though the two are usually set equal — which is also the one setting a workaround below silently changes what actually gets fetched at.

Whenever metrics.step >= metrics.rate_window (true at the usual-equal default), the collector does not send metrics.step to Prometheus as-is: a live-confirmed gigapipe bug returns zero series for any rate()-based range query or quantile_over_time subquery once the query's own step reaches the function's range-vector duration, so this tool queries at half that step (or the largest whole-second step below it that divides metrics.step evenly, at a wider ratio) and reassembles the configured grid from the result. Every disk's coverage, load history and forecast input comes back numerically identical to what a plain metrics.step query would have returned on an unaffected backend — except the quantile_over_time decision statistic itself, whose inner evaluation runs on the denser grid unconditionally, on every backend (a small, real shift in the reduced 95th-percentile value; see docs/internals/30-metrics.md). The *stored* sample count this produces — what collect-testdata --estimate prints and support.max_series_points compares against — is correspondingly doubled (or more) at the same ratio.

metrics.pvestatd_push_interval

Duration, default 10s (pvestatd's own default).

How often PVE's pvestatd pushes metrics — a PVE-side setting this tool cannot read from the API, so it must be declared here. verify-metrics cross-checks it against the *observed* sample spacing of a live series and errors if they disagree by more than 20%, which is what stops a stale declaration from silently invalidating the rate_window rule above.

window — the decision window

This is the period whose load is balanced. It is **not** the amount of history a forecast needs to fit — see forecast.model below and IMPLEMENTATION_PLAN.md section 10.1.

window.lookback

Duration, default 24h.

The window each disk's load is reduced over: the last window.lookback under forecast.model: quantile, the next one (as forecast) under holt_winters. Must be at least as long as the configured forecast.model needs (forecast.model: holt_winters at its default seasonal_periods needs 48h, which a 24h lookback can never supply — this is rejected at startup, not silently degraded).

window.quantile

Fraction in (0, 1), default 0.95.

The point-estimate quantile: each disk's load is the p95 of its raw signal over window.lookback, not the mean, so one traffic spike neither triggers nor suppresses a migration.

window.min_coverage

Fraction in (0, 1], default 0.80.

A disk whose sample coverage over window.lookback falls below this fraction has its data rejected for this run; its last known load from state.json is used instead, flagged in the report. Never treated as zero — that would silently invite migrations *onto* a busy but under-sampled storage.

`load_weights` — combining read/write and time/ops/bytes

Each disk's load is a single scalar, in **average in-flight I/O requests**, blended from three raw per-disk signals — I/O time, operations/second and bytes/second — after applying the read/write asymmetry factors below. The three raw signals have wildly different magnitudes (in-flight I/O is roughly 0–10, ops/s can run into the tens of thousands, bytes/s into the billions), so they cannot be weighted directly: each is first normalized against its own group's total (which makes the three comparable, but throws away their physical meaning), the normalized values are blended using the weights below, and the blend is then rescaled back onto the in-flight-I/O scale using the group's total I/O time. Under the defaults (`iotime: 1.0`, `ops: 0.0`, `bytes: 0.0`) that rescale is an exact identity, so a disk's load is simply its own I/O-time term — the normalize/blend/rescale machinery only changes the result once `ops` or `bytes` carries a non-zero weight. The absolute scale this produces matters beyond the blend itself: `migration.source_load_weight` and its siblings below assume that `1.0` in-flight request at a mirror endpoint is directly comparable to a disk's own load, which is only true because of this rescale. (Derived in full, with the normalization formula, in `IMPLEMENTATION_PLAN.md` section 4.)

`load_weights.iotime`

Weight, default `1.0`.

Weight on the I/O-time term (`rate(rd_total_time_ns + wr_total_time_ns) / 1e9`), the primary signal by default: it is Little's-law in-flight I/O, additive across disks and self-weighting between large sequential and small random access with no manual read/write tuning needed.

`load_weights.ops`

Weight, default `0.0`.

Weight on the operations/second term. Non-zero only if you want IOPS to influence placement independently of I/O time.

`load_weights.bytes`

Weight, default `0.0`.

Weight on the bytes/second term, for the same reason as `load_weights.ops`.

`load_weights.read_factor`

Weight, default `1.0`.

Read/write asymmetry applied to the `ops`/`bytes` terms only (see `load_weights.write_factor`). Applying it to the `iotime` term as well double-counts, since I/O time already reflects the real read/write cost difference — do that only deliberately.

`load_weights.write_factor`

Weight, default `1.0`.

As `load_weights.read_factor`, for the write side. A write-heavy RAID-6 array, for example, might set this above 1.0 on the `ops`/`bytes` terms to express the real cost asymmetry those terms cannot otherwise see.

`groups` — storage groups

`groups[].name`

String, required.

A group's name. A disk may only ever move between storages in its own group; there is no cross-group balancing (`IMPLEMENTATION_PLAN.md` section 5). Must be unique.

groups[].storages[].id

String, required.

A PVE storage id, or a `/regex/` pattern — a value that both begins and ends with `/` — matching one or more storage ids (`IMPLEMENTATION_PLAN.md` section 11.4). A pattern is matched with `re.fullmatch` (case-sensitive) against the live cluster's storage inventory on every run, so `/san-.*/` picks up a LUN added after the config was written with no edit needed; its own entry's `capability_weight`/`reserve_factor` apply to every storage it matches. A literal entry always overrides a pattern that also matches its storage, so one member of a pattern-matched family can still be pinned to different options. A storage may belong to **at most one** group (after pattern expansion) — belonging to two would make a disk's eligible destinations ambiguous, and config/cluster validation rejects it, as does a pattern matching zero storages or two patterns in one group claiming the same storage.

groups[].storages[].capability_weight

Weight, `> 0`, default `1.0`.

The storage's **relative** share of the group's load — `u_s = L_s / c_s` is what actually gets equalized, so a storage with half the spindles of its peers can be given `0.5` and will correctly be assigned half the load. Only the *ratios* within a group matter; the absolute value is meaningless and it is not a capacity.

groups[].storages[].reserve_factor

Weight or `null`, default `null` (inherits `snapshot_reserve.factor`).

Per-storage override of the group-wide snapshot reserve factor, for a storage whose snapshot behaviour genuinely differs from its peers.

groups[].storages[].enforce_format

`raw`, `qcow2` or `null`, default `null` (keep the disk's current format).

Normally a disk keeps its format when it moves: a `qcow2` disk lands as `qcow2`, a raw one as raw, and a storage that cannot hold that format is never chosen as a destination. Setting `enforce_format` on a storage changes that for **moves onto it**: every disk the plan moves there arrives in the named format, converted during the move if it was in another one. The typical use is shared (non-thin) LVM on Proxmox VE 9.2, where a `qcow2` volume is what makes volume-chain snapshots possible: `enforce_format: qcow2` makes every disk that lands on that LUN snapshot-capable; `raw` does the opposite, for the speed of a plain block device. It can be set on a literal entry or on a `/…/` pattern entry (every storage the pattern matches gets it; a literal entry for one of them overrides the pattern wholesale, as for every per-storage option). There is no global setting, because which formats exist depends on the storage type. `vmdk` is not offered.

What it does *not* do:

- It never causes a move. A disk already sitting on the storage in another format is left alone (`verify-storages` counts such disks so you can see how far the storage is from its policy); converting those in place is something you do with `qm disk move` yourself. A disk is converted only as a side effect of a move the balancer wanted for its usual reasons.
- It never touches a VM's TPM state disk (`tpmstate0`), which always keeps its format, and a disk with snapshots is never moved at all.
- It must fit the storage's type: `qcow2` is accepted on `lvm`, `dir`, `nfs`, `cifs` and `cephfs` storages, `raw` everywhere. A value the storage's type cannot hold is refused when the cluster is read, naming the storage — which for a pattern entry means "narrow the pattern".

Space accounting. A disk that is converted is counted on the target at the larger of its current size and the `size=` in its VM configuration, and, for a `qcow2` volume on an `lvm` storage, plus the image's own metadata (a small fraction of a percent; the tool uses a safe upper bound). That larger figure is what the snapshot

reserve, your free_space requirement and the in-flight space check are computed with, so a conversion cannot quietly eat into them. Just before a converting move starts, the tool re-reads the volume; if its format is no longer the planned one the move is skipped and the next run plans again.

Too much policy has a price: qcow2 → raw on LVM gives up volume-chain snapshots on that storage for the moved disks, and raw → qcow2 adds a qcow2 layer to every I/O. Leave it null unless you want one of those outcomes. The move line of plan shows a conversion as raw→qcow2.

groups[].storages[].free_space.soft / groups[].storages[].free_space.hard

Size, byte-unit string, percentage string ("N%"), or null — each independently. null here means *inherit the global free_space.soft/.hard value below* (§free_space — the same per-storage-null-means -inherit rule reserve_factor above already has). One member of a /…/-matched family can still be pinned to a different requirement than its peers, the same way a literal entry already overrides a pattern's reserve_factor.

A percentage is resolved against **this storage's own capacity**, even when it comes from a pattern entry matching several differently-sized storages — "10%" demands different byte counts on a 20 TiB and a 2 TiB LUN, which is the point: "a tenth of the LUN free" is one policy applied per storage, not one number shared across the family.

snapshot_reserve — the free-space floor

snapshot_reserve.factor

Weight >= 0, default 2.0.

Keep this many times the **largest VM footprint** on a storage free at all times, including *during* a migration, not merely before and after. A snapshot is taken of a whole VM — every one of its disks at once — and PVE 9's volume-chain snapshots allocate a new full-size volume per snapshotted disk. So what a snapshot of VM *v* needs on storage *s* is the sum of all of *v*'s disks that live on *s* (its *footprint* there), not its largest single disk: a VM with five 2 TiB disks on one storage needs 10 TiB free for one snapshot, and at the default 2.0 the storage keeps 20 TiB free for it. The footprint is counted per storage — the room a snapshot needs on *another* storage holding some of the same VM's disks is that storage's own footprint figure. Pinned disks count (they are snapshotted with their VM); disks of stopped VMs count toward used space but carry no footprint, because stopped VMs are not planned for snapshots. show-load and verify-storages print the largest footprint on each storage and the VM it belongs to. This is the constraint the tool never trades against balance.

Upgrading from a release that reserved only the largest disk: any storage holding two or more disks of one VM now reserves more than before, and one that was compliant can report a reserve shortfall it did not have yesterday. That is the rule being right, not the cluster changing, and the tool treats it as a repair like any other shortfall — the first run after the upgrade can plan substantial migrations. Read show-load (and a dry-run plan) first.

snapshot_reserve.split_vm_footprint

Size, byte-unit string, or null, default 2TiB.

The most of **one VM** that **one storage** should hold. A VM whose disks add up to more than this is spread over several storages until no storage holds more than this much of it — or as little as the storages available allow. With fewer storages than disks, which is the usual case, the cap is not reachable and need not be: five 2 TiB disks on three storages end up 2 + 2 + 1 disks (4 + 4 + 2 TiB), the smallest peak three storages allow, and the fullest storage then needs 2.0 × 4 TiB = 8 TiB free for a snapshot instead of 20 TiB. A single disk larger than the cap cannot be split, so its own storage may hold it in full; only what is *added* to it there counts as excess. The rule is a standing preference, not a repair: it never displaces a reserve or free-space repair, and it never creates a shortfall. objective.mu_vm_split_per_tib sets how strongly

it is pursued; a group holding such a VM on one storage is planned even when its I/O and data are in balance. `null` turns the rule, that weight and the planning trigger off. Must be `null` or above `0` (a `0` would spread every multi-disk VM — a typo, not a policy). **Too low** spreads many VMs for little gain and costs migrations; **too high** keeps large VMs together and leaves their snapshot reserve large.

`snapshot_reserve.count_foreign_volumes`

Boolean, default `true`.

Count volumes DRS does not manage (templates, ISOs, backups, other groups' disks, orphans) against a storage's used capacity. Strongly recommended: turning this off understates real usage and silently erodes the reserve it is meant to guarantee.

`free_space` — keep N bytes (or N%) free on top of the snapshot reserve

The Storage-DRS mandate (`IMPLEMENTATION_PLAN.md` section 5.3.1): an admin places a new VM on a storage this tool does not manage the free space of by snapshot behaviour alone, and the engine migrates data off it until the configured free space is free again — bypassing the drift/imbalance gates and the payback economics for exactly this reason, the same way a snapshot -reserve violation already does. `soft` and `hard` here are the *global* defaults; `groups[].storages[].free_space.soft/.hard` above override them per storage or per `/…/`-matched pattern, most specific wins.

`free_space.soft`

Size, byte-unit string, or percentage string (`"N%"`, `0 <= N < 100`), default `0`.

The **plan-endpoint** requirement: the number of bytes that must be free on a storage once the plan has fully run. `R_s = max(factor * largest_vm_footprint, soft)` — the *larger* of the snapshot term and this floor wins, on every storage, always; neither term can erode the other. The default `0` changes nothing for a config that sets no knob in this block at all: the resulting model is identical to the pre-`free_space` snapshot-only floor. A storage that ends the plan below its `soft` requirement is in violation exactly like a snapshot-reserve breach, and the engine migrates disks off it until the requirement is met or reports the residual shortfall as unfixable (`plan`'s `⚠ unfixable shortfall` line, with the amount and the pinned disks in the way) — it does not guarantee the requirement is *achievable*, only that it is pursued unconditionally.

`free_space.hard`

Size, byte-unit string, percentage string, or `null`, default `null` (= `soft`: no dip at all).

The **transient** floor: how far a storage may dip below `soft` while a disk is landing on it mid-migration (section 8.1) — a mirror target is fully allocated before its source releases anything, so a plan that lands a disk on a storage that is *heading toward* `soft` needs room to pass through it without ever crossing `hard`. The default `null` keeps the transient invariant exactly as strong as it was before `free_space` existed: no dip below `soft` at all. Set it strictly below `soft` only when you have deliberately decided to trade a bounded, planned dip for it — a transient dip can only ever land on a storage the *finished* plan leaves compliant, never used to permanently weaken the requirement. `hard` above `soft` is a config error: a floor stronger than the requirement it is supposed to relax would make every plan for an already-compliant storage infeasible.

`gates` — deciding whether to act at all

`gates.drift_threshold`

Fraction, default `0.10`.

Re-planning is skipped until the load vector has moved by this much in L1, relative to the vector recorded at the last executed balance. Lower reacts sooner and migrates more; below roughly 0.05 the tool will start chasing noise on a busy cluster. Interacts with `gates.imbalance_threshold`: drift decides *whether to look*, imbalance decides *whether to act*.

The vector compared is the load the balancer itself acts on, whatever produced it: with `forecast.model: holt_winters` that is the forecast-scaled load, and the vector recorded at the last balance is the forecast-scaled one too. A change in forecast state between two runs — switching `forecast.model`, or the backtest starting to pass or fail — therefore counts as drift even when the workload did not change, by design: the picture the balancer is placing disks against changed, so reconsidering the placement is correct. It only opens the imbalance gate; a plan still has to clear the payback test.

`gates.imbalance_threshold`

Fraction, default `0.20`.

The minimum relative spread across a group's storages before a plan is actually built. A group under this threshold is left alone even if it has drifted.

`gates.capacity_spread_threshold`

Fraction or `null`, default `0.25`.

The data-spread counterpart of `gates.imbalance_threshold`: the minimum relative spread in fill fraction (bytes used / capacity) across a group's storages before a plan is built, evaluated even when the group's I/O is already perfectly balanced – in that case `objective.delta_capacity_spread` is what does the spreading. It can legitimately exceed `1` (a group filled to 5% that keeps 60% of its bytes on one storage deviates as much as a full one does). Unlike the drift/imbalance gates, this one **bypasses** them when it fires – a stable, balanced workload is not a reason to keep data concentrated on one storage. `null` disables the gate outright; the gate also cannot fire on a group with no data at all (mean fill `0`).

Opening a group is not the same as moving data. When this gate fires, the group is planned; whether anything moves is decided by the objective, and on a group whose I/O is already balanced only `objective.delta_capacity_spread` can pay for a move. That weight is a price per TiB, so a disk pays for its own move once it is larger than a break-even size (about 270 GiB at the defaults; the entry for that weight derives it). When the only disks that would help are smaller, `plan` reports `→ ACT` followed by `no moves made`, naming `objective.beta_move_count`, `objective.gamma_move_bytes_per_tib` and `objective.delta_capacity_spread` and printing the break-even size. Either raise `objective.delta_capacity_spread` (or lower `objective.beta_move_count`) until the moves you want pay, or raise this threshold (or set it to `null`) if you do not want groups planned for data spread at all. The gate itself stays relative on purpose: it asks whether the data is lopsided enough to plan for, which is a question about proportions, whereas the objective asks whether a particular move is worth it, which is a question about bytes.

`gates.cooldown_per_disk`

Duration, default `24h`.

A disk that moved within this long is pinned in place for planning purposes. Prevents a disk from being shuffled back and forth across two still-noisy storages.

`gates.cooldown_per_storage`

Duration, default `1h`.

A storage involved in a migration within this long accepts no new incoming moves. **Must exceed the implied wipe time of that storage's largest disk** (`largest_disk / |saferemove_throughput|` — the magnitude, see `verify-storages` on why that value is often negative) or the next run will plan onto a storage that is still draining and stall — `pve-storage-drs verify-storages` computes this and warns when the configured cooldown is too short.

`migration` — cost, bandwidth and the payback rule

`migration.bwlimit_bytes_per_sec`

Size/s, default `209715200` (200 MiB/s).

Passed to `move_disk` as `bwlimit` (converted to KiB/s at the API call site — the API's own unit, never bytes/s). Also the divisor in the mirror-duration estimate the payback test uses: a move's mirror is assumed to take `disk_bytes / bwlimit_bytes_per_sec` seconds (see `migration.payback_ratio` below for the full cost/benefit comparison).

This is the **whole** throttle on a migration: it may start at any time, and the tool does not model storage saturation. The other limits are `migration.max_single_move_duration` and the transient reserve invariant (see `snapshot_reserve.factor`).

`migration.source_load_weight`

Weight, default `1.0`.

Additional in-flight I/O charged to the **source** storage while a mirror is running — a mirror is one sequential reader, so `1.0` is the natural value in the same units as the load model.

`migration.target_load_weight`

Weight, default `1.0`.

As `migration.source_load_weight`, for the target (one sequential writer).

`migration.payback_horizon`

Duration, default `365d`.

The horizon `H` over which a plan's imbalance, data-spread and VM-affinity reduction are assumed to persist — `benefit = (alpha_spread * ΔE + delta_capacity_spread * ΔF + kappa_vm_affinity * ΔA) * H`, where each `Δ` is the improvement the plan buys in that term (its value before the plan minus its value after: `ΔE` for imbalance, `ΔF` for data spread, `ΔA` for VM-affinity fragmentation — see `objective` below for what each term measures). `ΔA` may be negative (a balance move that splits a VM pays for that fragmentation out of its other gains — `IMPLEMENTATION_PLAN.md` section 7.2). Setting this to `0` disables the payback test entirely (`IMPLEMENTATION_PLAN.md` section 11.1 rejects that at config-load time: `payback_horizon > 0` is required). This is an explicit assumption about how long a placement lasts, not about operator patience: a migration's cost is paid once and early, while its benefit accrues for as long as the workload keeps running on the new placement — infrastructure timescales are months to years, and the default assumes at least one year. Configuration validation warns (does not error) below `30d`, where the test starts rejecting real, slow-accruing benefit again; lower it toward the actual lifetime of your VMs only for genuinely short-lived fleets (CI runners, render farms, lab clusters).

`migration.payback_ratio`

Weight `> 0`, default `10.0`.

`λ`: a plan is only accepted when `benefit >= payback_ratio * cost`, summed over every disk the plan moves. Both sides are in **load-seconds** — average in-flight I/O requests (the unit `load_weights` produces, see `metrics.read_time_ns` above) multiplied by seconds — which is what makes the comparison meaningful. A disk's cost (`cost_d`) is the extra in-flight I/O the migration itself imposes: `migration.source_load_weight` on the source and `migration.target_load_weight` on the target for the mirror, which takes `disk_bytes / migration.bwlimit_bytes_per_sec` seconds, plus — when `migration.account_saferemove_wipe` is on and the source storage has `saferemove` set — `migration.wipe_load_weight` on the source for as long as the old volume takes to be zeroed. Benefit is the improvement the plan buys in imbalance, data spread and VM affinity, each weighted exactly as in the `objective` section below and held for

migration.payback_horizon (see above for that formula). This is the numeric form of "migrating a very large disk might generate more traffic than it saves." (IMPLEMENTATION_PLAN.md section 7.2.)

migration.max_single_move_duration

Duration, default 6h.

A hard per-move ceiling on duration_mirror + duration_wipe. A move predicted to take longer than this is rejected outright regardless of the aggregate payback test.

migration.account_saferemove_wipe

Boolean, default true.

Include the post-move saferemove zeroing pass in cost and duration estimates. PVE's LVM saferemove defaults to 10 MiB/s, so a 1.5 TiB disk's cleanup (~44h) can dwarf its ~2.2h mirror; disabling this only if you have verified saferemove is off on every storage in a group understates real migration cost badly if that assumption is wrong.

migration.wipe_load_weight

Weight, default 1.0.

In-flight I/O charged to the source for the whole saferemove wipe duration in the cost model. The zeroing pass is one sequential writer, so 1.0 is the natural value.

migration.assume_thick_provisioning

Boolean, only true (the default) is accepted. **Kept so a config that spells it out still loads; false is refused at startup.**

This tool never considers over-provisioning. Every disk counts at its *provisioned* size — in the reserve arithmetic, the free-space requirement, the transient check and the cost model — on a thin-provisioned storage (Ceph RBD, LVM-thin, ZFS) exactly as on a thick one. Thin provisioning is what lets a pool hold more provisioned bytes than it has; here it is deliberately never counted on, because a move that only fits *if the disks stay thin* is one a growing guest can turn into a full pool.

The consequence to plan for: on a thin pool the tool's idea of "used" is the sum of the disks' sizes plus foreign volumes, which can be several times what the pool reports as allocated. A free_space.soft of "90%" on a pool with plenty of *actually* free space can therefore show a large shortfall in plan/explain, and the engine will migrate disks off it. That is the intended behaviour, not a miscalculation.

The same holds at the moment of the move: the check apply makes just before each move_disk re-reads the target's volume listing and sums the volumes' provisioned sizes, never the pool's allocated figure, so it can refuse a move onto a thin pool that still looks half empty in the PVE UI. show-load, plan and explain print the provisioned figure as the storage's usage and, when the pool's own number differs, add it in brackets as (pool reports X allocated).

migration.tiny_disk_bytes

Size, default 67108864 (64 MiB).

A disk smaller than this carries zero beta_move_count/gamma_move_bytes_per_tib cost in the objective and zero cost_d in the payback model — it still counts as a scheduled move and every hard per-move safety rule (max_single_move_duration, the transient reserve invariant) still applies to it exactly like any other move, but it needs no payback verdict and cannot make a plan fail the aggregate payback_ratio test. Together with objective.kappa_vm_affinity, this is what lets a tiny volume — an efidisk0 var store or tpmstate0, both normally a few hundred KiB to a few MiB — rejoin its VM for free instead of being priced like a real migration: both device types move online on PVE 9.2, so there is no reason to price their

reunion like a real migration once they are cheap enough to ignore. (IMPLEMENTATION_PLAN.md §§5.4, 7.1, 7.3, 3.6.)

The default sits comfortably above either of those and far below any disk the payback rule was written for, so a 528 KiB EFI disk always qualifies and a 64 GiB data disk never does. Set it to 0 to restore the pre-section-7.2 accounting, where every disk — however small — is charged a full migration. Setting it too high (larger than disks you actually want cost-accounted) lets real, meaningfully-sized migrations bypass the payback safety test entirely; there is no upper bound enforced beyond ≥ 0, so this is an operator judgement call, not a validated range.

A disk below this size only ever moves together with its VM. It may end a plan where it is now, or on a storage where at least one *larger* disk of the same VM, in the same storage group, ends the plan — never on a storage of its own. So an EFI or TPM disk follows its VM's system disk when that disk moves, rejoins it when the VM is already split, and otherwise stays put. Moving it alone would buy nothing: it carries no load worth balancing, and as a free-space "repair" it frees a fraction of a MiB while splitting the VM from its firmware state. Larger disks are not bound this way: a VM's big disks can still be placed on different storages when they do not fit together or when only a split balances the group (objective.kappa_vm_affinity is the soft preference against that). A VM whose disks in this group are *all* below the threshold — typically because its system disk is in a different group — has nothing to follow, and those disks are pinned with small disk with no larger disk of its VM in this group (migration.tiny_disk_bytes). Nothing ever moves between groups. Setting the threshold to 0 switches this rule off along with the cost exemption.

objective — the solver's trade-off weights

The solver (and, for the terms it also uses, the heuristic backend) minimizes one weighted sum, evaluated per group:

```
  alpha_spread            * (imbalance: summed deviation of each storage's utilization from the group's
  target)
+ beta_move_count         * (number of migrations)
+ gamma_move_bytes_per_tib * (TiB actually migrated)
+ kappa_vm_affinity       * (VM disk fragmentation, I/O-weighted — see below)
+ delta_capacity_spread   * (data spread: TiB by which each storage's data is off its capacity-proportional
share, summed)
+ mu_vm_split_per_tib     * (split excess: TiB of each large VM beyond the cap on its fullest storage)
+ reserve_violation_penalty * (reserve violation, heuristic backend only — see below)
```

The first six terms are calibrated to share one scale, in units of average in-flight I/O per storage (see metrics.read_time_ns above) — which is what makes the weights directly comparable: at the defaults, a migration must buy at least a 0.25-request reduction in summed imbalance just to cover its own beta_move_count cost, before its gamma_move_bytes_per_tib and kappa_vm_affinity costs and the separate payback test (migration.payback_ratio above) are even considered. Each term is explained, with its own default and worked numbers, in its own entry below. (IMPLEMENTATION_PLAN.md section 5.4 has the full derivation; section 14.3 works two competing plans through this exact formula.)

objective.spread_metric

l1 or minmax, default l1.

How imbalance is measured: l1 (sum of each storage's deviation from the group's target utilization) or minmax (only the single hottest storage). l1 is the default because minmax is indifferent to a second nearly-as-bad storage once the worst one is fixed.

objective.alpha_spread

Weight, default 1.0.

Weight on the imbalance term (the sum of each storage's deviation from the group's target utilization — see `objective` above for where this sits in the full objective, and why `1.0` against the default `beta_move_count: 0.25` means a migration must buy at least a 0.25-request reduction in that sum to be worth making at all).

`objective.beta_move_count`

Weight, default `0.25`.

Penalty per migration — the tunable implementation of "minimize the number of migrations" (`IMPLEMENTATION_PLAN.md`'s "Requirement interpretation" note: a strict minimum would refuse a second cheap move that halves the remaining imbalance, which is not desired). Raising it trades balance quality for fewer, larger moves.

`objective.gamma_move_bytes_per_tib`

Weight per TiB, default `0.05`.

Penalty per TiB actually migrated, biasing against moving large disks specifically (as opposed to `beta_move_count`, which only counts moves).

`objective.kappa_vm_affinity`

Weight, default `0.50`.

Penalty per extra storage a VM's disks are spread across, counted only **within** a group — a VM split across two groups is structural and cannot be repaired by any migration, so it is not counted — and weighted by the VM's own I/O share of the group: `w_v = max(1, ℓ_v / ℓ̄)`, where `ℓ_v` is the VM's total load (the sum, in average in-flight I/O, of every one of its disks in the group, including disks pinned this run) and `ℓ̄` is the group's mean load per VM (the group's total load divided by every VM that owns a disk in the group). The floor of `1` means a below-average VM's fragmentation is weighted exactly as configured; a VM running several times the group's average I/O is worth correspondingly more to keep together than the configured weight alone suggests — a VM doing 2.7x the group's average I/O, for example, is effectively weighted as if `kappa_vm_affinity` were 2.7 times higher for that VM alone. A soft preference: a strong imbalance or a capacity constraint can legitimately override it. (`IMPLEMENTATION_PLAN.md` section 5.4.)

`objective.mu_vm_split_per_tib`

Weight `>= 0`, default `1.0`.

Weight, per TiB, of the large-VM split rule (`snapshot_reserve.split_vm_footprint` above). For every VM above that threshold, the *excess* is how far its footprint on its **fullest** storage exceeds the VM's cap — the peak, not a sum over storages, because the reserve a storage needs is driven by the largest footprint on it. Moving a disk off the fullest storage lowers the excess by up to the disk's size and costs `beta_move_count + gamma_move_bytes_per_tib × TiB` (plus `kappa_vm_affinity` when it opens a storage the VM did not use yet), so at the defaults a move pays above about 0.26 TiB onto a storage the VM already uses and above about 0.79 TiB onto a new one; a VM made of many small disks splits more reluctantly than one with a few large disks. Raise the weight to split harder. A large VM's affinity weight is fixed at 1 (the I/O-weighting of `kappa_vm_affinity` does not apply to it), so a busy VM's I/O cannot veto the policy. `0` disables the term and the planning trigger; configuration validation warns when it is above `0` but below `kappa_vm_affinity`, where a TiB of excess no longer buys a new storage and the rule rarely acts.

`objective.delta_capacity_spread`

Weight per TiB, default `0.5`.

Price of one TiB of data spread. A storage's data spread is how far the data on it is from its capacity-proportional share of the group's data: if a group holds 20 TiB across storages of 30, 10 and 10 TiB, the shares are 12, 4 and 4 TiB, and a storage holding 15 TiB is 3 TiB off. The term is the sum of those deviations

over the storages, in TiB, times this weight. It bounds the share of a group's data any single storage failure costs, and keeps peak fill uniform across the group. `0` disables the term entirely.

It is absolute, so it does not depend on the group's size. Moving a TiB of data from a storage that has too much to one that has too little removes two TiB of deviation (one at each end) and so earns `2 × delta_capacity_spread` — `1.0` at the default — whether the group holds 5 TiB or 500. A disk of `X` TiB pays for its move when that gain exceeds its cost, `beta_move_count + gamma_move_bytes_per_tib × X`, which gives a break-even size

```
X* = beta_move_count / (2 × delta_capacity_spread − gamma_move_bytes_per_tib)  TiB
```

At the defaults that is `0.25 / (1.0 − 0.05) ≈ 0.263 TiB`, about 270 GiB: larger disks that sit on an over-full storage and have somewhere to go pay for their own move; smaller ones move only when something else — their VM's other disks, the I/O balance — moves them anyway. When `2 × delta_capacity_spread` is at most `gamma_move_bytes_per_tib` no disk of any size can pay, the term is on but inert, and configuration validation warns. `plan` prints `X*` in its `no moves made` line.

I/O balance stays the first priority, as an exchange rate: at the default, one TiB moved toward its share (a gain of `1.0`) is worth one unit of in-flight-load deviation (`alpha_spread` is per unit of in-flight load). On a group whose I/O is imbalanced, the I/O term dominates; on one whose I/O is already balanced, this term is what decides.

Choosing a value. Decide the smallest disk you want moved purely for data spread, `X_min`, and set `delta_capacity_spread = (beta_move_count / X_min + gamma_move_bytes_per_tib) / 2`. For `X_min = 100 GiB` (about 0.098 TiB) that is about `1.3`; for `X_min = 1 TiB` it is `0.15`. Too high moves many small disks for little gain and, because every TiB of spread now outweighs more of the I/O term, lets data evenness override I/O evenness; too low (or `0`) leaves data spread to chance: the capacity gate still opens groups, and nothing moves for it. **Upgrading:** earlier versions priced the deviation relative to the group's mean fill, so a value raised under that formula does not carry over — re-derive it from `X*`. A replayed bundle with five equal storages, balanced I/O and data concentrated on three of them planned no capacity move under the relative form at the default and plans 20 data-disk moves under the absolute one, from 100 GiB disks that follow their VM up to 1.95 TiB (observed on the operator's own cluster).

`objective.affinity_counts_pinned_disks`

Boolean, default `true`.

Whether a disk that cannot move this run (snapshot-blocked, excluded, or on a locked VM) still counts toward `kappa_vm_affinity`. Default `true`: such a disk still *occupies* a storage, so the VM really is spread, and its fixed location is an anchor the VM's movable disks can be drawn back to. Note `efidisk0`/`tpmstate0` are *not* in this pinned set on PVE 9.2 — they move online.

Set it to `false` to count movable disks only. Be aware of what that costs. Take a VM with a snapshot-blocked disk on `san-a`:

- **One movable disk, on `san-b`.** With `false` the VM's counted footprint is that one disk, so it is never "spread", and moving it to `san-a` (reuniting the VM) scores exactly the same as leaving it or sending it to a third storage — the reunion is invisible, and the solver may pick any of them. With `true` the VM counts as spread over two storages until the disk joins `san-a`, and `kappa_vm_affinity` pulls it there.
- **Two movable disks, both on `san-b`.** With `false` the counted footprint is `san-b` alone; moving one of them to `san-a` to rejoin the pinned disk makes it `{san-a, san-b}` and is charged `kappa_vm_affinity` as if the VM had just been split. With `true` that same move is neutral, and moving both is a gain.

The pinned disk's storage is fixed, so counting it never asks for anything unreachable: it only prefers targets where the VM already has a disk.

This default changed after 0.1.6 (it was `false`). If you have not set it and your cluster has pinned disks, expect plans to prefer targets where the VM already has a disk. To keep the old behaviour, set it explicitly.

`objective.reserve_violation_penalty`

Weight, default `1000.0`.

The reserve-violation weight in the **heuristic backend**'s objective (`solver.backend: heuristic`, or `auto` falling back to it when neither a MILP backend is installed): multiplied straight into `reserve_penalty_term = objective.reserve_violation_penalty * reserve_shortfall_tib`, one of the seven terms `explain`'s `objective:` line prints. It is used exactly as configured — no floor, no automatic raise, no warning.

The MILP backend (`solver.backend: cbc`, and `auto`) solves the reserve **lexicographically** instead: the reserve is fixed as a hard constraint and solved for first, before the rest of the objective above is even considered, so no weight — this one included — can trade it away. A single-stage alternative exists on paper: computing a provably-dominant penalty from the group's own load instead of taking this key at face value, with a `max(configured, computed)` floor. It is not implemented, so this key never gets raised automatically for any backend — what you set is exactly what the heuristic backend uses, and the MILP backend never consults it at all. (`IMPLEMENTATION_PLAN.md` section 5.3.)

`solver` — which backend plans

`solver.backend`

One of `auto`, `cbc`, `heuristic`; default `auto`.

`auto` uses CBC — the packaged solver path via `python3-pulp` + `coinor-cbc`, both hard dependencies of the Debian package — and falls back to the dependency-free heuristic if that library is not importable or its `cbc` binary cannot run (a plain `pip install` without the `solver` extra; never on a Debian install). `cbc` forces that one backend, failing that group's solve back to the heuristic if its library is not importable or it cannot solve within `solver.time_limit_seconds` – force a specific backend only to reproduce or compare a result. `heuristic` skips the solver entirely. See `docs/internals/91-optimize.md`.

(A `cpsat` value existed through release 0.1.8, selecting an optional `ortools`-based backend; it was removed — `ortools` has no Debian package and the backend could therefore never run on a PVE host — and a config that still names it now fails schema validation, loudly, as it should.)

`solver.time_limit_seconds`

Duration, default `60`.

Wall-clock budget for a MILP solve attempt before falling back to the heuristic. Never emits a partial/unvalidated assignment on timeout — it falls back cleanly instead.

`solver.mip_gap`

Fraction, default `0.02`.

Acceptable optimality gap for the MILP solve: CBC may stop once the best solution found is within this fraction of a proven lower bound, rather than solving to exact optimality. (`IMPLEMENTATION_PLAN.md` section 5.5.)

`solver.heuristic_iterations`

Integer `> 0`, default `5000`.

Iteration budget for the heuristic's local-search descent phase. Higher can find a better local optimum on a large group at the cost of run time. (`IMPLEMENTATION_PLAN.md` section 5.5.)

`execution` — how (and whether) moves actually happen

`execution.mode`

One of `dry-run`, `confirm`, `auto`; default `dry-run`.

`dry-run` (the default) prints the plan and changes nothing. `confirm` executes one move at a time with a prompt before each. `auto` executes unattended, subject to the concurrency and time-window settings below. `--mode` on the command line overrides this for one run; moving *toward* less safety is logged at warning level (`IMPLEMENTATION_PLAN.md` section 11.3).

`execution.max_concurrent_migrations`

Integer `>= 1`, default `1`.

How many moves may be in flight across the whole run at once, in `--mode auto` only (`dry-run`/`confirm` always run strictly sequentially regardless of this setting). Above `1` requires the *generalized* form of the transient reserve invariant (see `snapshot_reserve.factor` above for the single-move definition): several disks can land on one storage at once, and none of their sources release space until each individually completes — `apply` re-checks it live before launching each move. Launch order stays strictly FIFO: `apply` never reorders the scheduler's own queue to keep every slot busy, so a move that cannot launch yet is waited for rather than skipped past — see `docs/manual/28-apply.md`'s own "Concurrent execution" section.

`execution.max_migrations_per_run`

Integer `>= 1`, default `5`.

A ceiling on how many moves one invocation executes, independent of how many the plan contains — the remainder waits for the next run. `auto` mode only (`dry-run`/`confirm` execute or report the whole plan regardless); a shared budget across every group the run visits, not reset per group.

`execution.max_concurrent_per_storage`

Integer `>= 1`, default `1`.

Caps concurrent moves touching one storage, counting it as either source or target, in `--mode auto` only. At the default of `1`, two moves targeting the same storage serialize automatically and the generalized transient invariant collapses to its simple single-move form. Raising it should be a deliberate act on a storage with real spare headroom — two moves off the same source also means two concurrent `saferemove` wipes sharing one throttle.

`execution.max_replans_per_run`

Integer `>= 0`, default `3`.

A cap on how many times one run may abandon its current plan and re-plan from newly observed state — a mismatch between the plan and reality (a VM live-migrated mid-plan, a lock appeared, a disk created or moved by someone else, another tool filling the target) is normal, but a cluster churning faster than the engine can plan is a condition for a human, not for indefinite retrying. `auto` mode only — `confirm`/`dry-run` report a mismatch (`replan_needed`) and stop that group's own run for the operator to re-run by hand, rather than re-planning automatically.

When the cap is reached the run bails out and exits `0`; the next run starts from freshly observed state. Set too low, ordinary churn ends runs early and moves wait for the next one; set too high, a cluster that keeps changing under the tool keeps it planning instead of waiting. `0` disables re-planning: the first mismatch ends the run. Only a *mismatch* re-plans — an error reading PVE or Prometheus fails the run outright and never counts against this cap (see `28-apply.md`, "When something cannot be read").

execution.abort_on_failure

Boolean, default `true`.

On a failed move, stop the run and report rather than continuing with the remaining plan. The cluster is left in a valid intermediate state either way; the next run re-plans from reality.

execution.poll_interval_seconds

Duration, default `10`.

How often an in-flight `move_disk` task's status is polled.

execution.api_outage_timeout

Duration, default `1h`. `0` disables.

While `apply` executes, a Proxmox VE API read that fails because the API cannot be reached — connection refused or timed out, a TLS handshake failure, an answer that is not valid API output (such as a firewall's error page), or an HTTP 429 or 5xx, as during network maintenance or a node restart — is retried with back-off (5 s doubling to 60 s) for up to this long, instead of ending the run while a migration may still be going on. The clock is per call and restarts after each success. An error the API actually answers (4xx, in particular 401/403 "not authorized") is never retried.

Outside this window, every read still gets a short retry (about 10 s in total: 1 s, 2 s, 4 s, with jitter), so a single dropped connection or a virtual IP moving to another node does not fail a dry run or `show-load`. Each retry after a connection-level failure starts from a fresh login and connection. Failover between API hosts is not done here: point `proxmox.host` at a name or virtual IP that already fails over. `move_disk` itself is never retried blindly: a connection that dropped after the request went out may still have started the task; if the run does give up, the next run discovers a still-running move (`state.json` and the cluster task list) and leaves that VM alone.

execution.locks.wait_timeout

Duration, default `4h`.

How long to wait for a VM's config `lock` (any non-empty value — the set is treated as open-ended and never whitelisted) to clear, or for a task still running on the VM (such as the "Erase data" job of an earlier move) to finish, before applying `execution.locks.on_timeout`. Generous by default: a backup of a large VM can easily run for hours, and that is a normal, not exceptional, condition.

execution.locks.poll_interval

Duration, default `30s`.

How often the lock is re-checked while waiting.

execution.locks.on_timeout

One of `skip`, `abort`; default `skip`.

What happens when `execution.locks.wait_timeout` elapses: skip this move and continue with the rest of the plan, or abort the run. Never "force" — that option does not exist, deliberately.

execution.locks.task_retry_limit

Integer, default `2`.

The `move_disk` *task itself* can fail immediately with PVE's own `can't lock file '/var/lock/qemu-server/lock-<vmid>.conf' - got timeout` — a different lock than the config `lock` attribute `execution.locks.wait_timeout` waits out, momentarily still held (typically by the previous move's own `saferemove` wipe/cleanup finishing for the same VM) even after that attribute already reads clear. This

many extra attempts are made, each preceded by execution.locks.task_retry_backoff, before the move is reported failed like any other task failure. 0 disables the retry.

execution.locks.task_retry_backoff

Duration, default 15s.

How long to wait between a task-lock-timeout failure and the next move_disk retry attempt.

execution.source_release.wait

Boolean, default true.

Wait for the source volume of a completed move to actually disappear (and the VM to be unlocked) before considering the move done, rather than trusting the move_disk task's own OK status — with saferemove enabled, OK does not mean the source space is back yet. Set false only on storages verified not to wipe.

execution.source_release.timeout

Duration, default 48h.

Bound on the wait above, sized for a multi-TiB disk wiping at the default 10 MiB/s. Must be at least largest_disk / |saferemove_throughput| for every storage where saferemove is on — verify-storages checks this.

execution.time_windows[].days

List of mon..sun, default: every day.

Which days a time window applies on. Applies only to execution.mode: auto — dry-run and confirm are not time-restricted, since neither changes anything without a human already present.

execution.time_windows[].start / execution.time_windows[].end

HH:MM strings, required together, must differ.

The window auto mode is permitted to execute moves in, in the **local time of the host running pve-storage-drs** — the same convention systemd.timer's own OnCalendar= uses by default — not UTC. A window may cross midnight (start: "22:00", end: "06:00"); start == end is rejected as ambiguous — it would silently mean either "never" or "always". No time_windows configured at all means no restriction: auto may execute at any time — this is the deliberate default, not merely the absence of a rule. Before starting a move, auto also refuses it (and stops the group's run cleanly, without aborting a move already in progress) if its estimated duration would not finish before the window closes.

exclude — what DRS never touches

exclude.vmids

List of integers, default [].

VM ids whose disks are never moved.

exclude.disks

List of "vmid:device" strings, default [], e.g. ["101:scsi1"].

Individual disks excluded regardless of their VM.

exclude.storages

List of storage ids, default [].

Storages excluded as both source and target — effectively removes them from their group without editing groups.

`exclude.tags`

List of PVE tags, default `["no-drs"]`.

A VM carrying any of these tags opts out entirely.

`exclude.skip_vms_with_snapshots`

Boolean, default `true`.

Pre-filter disks with a snapshot or an unreferenced companion volume before they reach the API. Setting this `false` does **not** make such a move work — PVE rejects `move_disk delete=1` on a snapshotted volume regardless; it only removes the early, informative warning. Keep this `true`.

`exclude.stopped`

Boolean, default `true`, and `true` is the only accepted value.

Stopped VMs are never touched: whatever violation their disks cause – a storage below its snapshot reserve, below the free space you configured, or badly out of balance – the tool does not move a stopped VM's disks. This is a safety property rather than a tuning knob. Replication and backup products such as Veeam track a VM's volumes by their location, and moving the disks of a replicated VM breaks the replica badly.

A VM whose status is not `running` is skipped before its configuration is even fetched. Its disks still occupy space on their storage, so their bytes are counted as foreign volumes and reduce the room the planner has to work with; the planner then repairs a violation by moving running VMs' disks instead, or reports that it cannot. A VM that was running when the plan was made but has stopped by the time its move is due is refused at apply time and the run asks for a re-plan.

The key exists only so that a configuration which spells it out keeps loading. `false` is **refused at load time** with an error naming the setting, because it would promise a mode that does not exist. There is no too-high or too-low: the setting has one value.

Renamed from `running_only`, which meant the same as `exclude.stopped: true`. A configuration still carrying the old key is rejected at load time as an unknown key; remove it.

`exclude.include_unused_disks`

Boolean, default `true`.

Whether detached (`unusedN`) volumes are movable. They carry zero load, so the solver only relocates one to repair a capacity violation — never for balance, since there is no benefit to buy. Set `false` to pin them in place entirely; they still count via `snapshot_reserve.count_foreign_volumes` either way.

`report`

`report.warn_pinned_load_fraction`

Fraction in (`0`, `1`], default `0.25`.

If pinned load (snapshots, locks, exclusions) exceeds this fraction of a group's total, the residual imbalance may be structural rather than a planning failure, and `explain` says so alongside the best achievable spread given the pins — see `29-explain.md`.

`state`

`state.path`

File path, default `/var/lib/pve-storage-drs/state.json`.

Local disk, one copy per host, deliberately never /etc/pve — see 00-installation.md. Losing this file is safe but not free: cooldowns and the drift baseline reset, so the next run may migrate sooner than intended. Treat it as state to back up, not as a cache.

show-load and plan both read this file (if present) for the drift gate's history; neither writes it, and a missing or unreadable file just means every group evaluates as if it had never been balanced before — see ../internals/15-state.md.

forecast — placing disks for the load they will have

See IMPLEMENTATION_PLAN.md sections 10 and 12.1 and proxmox_storage_drs/forecast.py.

forecast.model

One of quantile, holt_winters; default quantile.

quantile: each disk's load is the p95 (window.quantile) of its I/O over the **last** window.lookback. It needs no fitting, no extra history and no optional package, and it is what every plan used before holt_winters existed; with it nothing below applies and no extra Prometheus query is made.

holt_winters: each disk's load is scaled to a Holt-Winters forecast of that same p95 over the **next** window.lookback — fit the disk's history, forecast window.lookback ahead, take window.quantile of the forecast path. The scaling is a ratio (forecast p95 ÷ the observed p95 of the same history), so the load stays on the scale the rest of the model is calibrated against; a disk whose load is rising, or that has a daily peak the last window happened to miss, is placed for what it is about to do. A disk that is flagged for low sample coverage, has no observed load, or whose model cannot be fitted keeps its observed load. Gates, solver, payback and ordering all see the same forecast load.

When holt_winters is worth selecting: a load that follows a daily (or weekly, see seasonal_periods) cycle, or that trends. On a load with no structure — noise around a constant — it has nothing to predict, and the gate below will say so.

The gate: it must beat doing nothing clever. Once per group, the model is fit on the older half of its own recent history ([now − 2·lookback, now − lookback)) and both it and the quantile model predict the p95 of the newer half; holt_winters is used for the group only if its error is no larger than the quantile model's. There is no threshold to tune. If it loses — or there is not yet 2 · window.lookback of history to check — the group uses quantile for that run, logged at warning. explain and plan --json show what happened per group: which model was used, both backtest errors, and how many disks were scaled or kept.

Requirements. window.lookback must span two seasonal cycles (2 · seasonal_periods · metrics.step; 48h at the defaults), which pve-storage-drs enforces at startup rather than silently degrading; Prometheus must hold 2 · window.lookback of history for the gate to run at all; and python3-statsmodels must be installed (without it every disk keeps its observed load).

forecast.holt_winters.seasonal_periods

Integer > 0, default 288 (24h at a 5-minute step).

Samples per seasonal cycle. Combined with metrics.step, this determines holt_winters's required history: 2 * seasonal_periods * metrics.step — 48h at the defaults, which a 24h window.lookback can never satisfy.

To fit **weekly** seasonality instead of daily (e.g. weekends look different from weekdays), widen this together with window.lookback: seasonal_periods: 2016 (7d at the default 5m step) needs window.lookback of at least 28 days (2 * 2016 * 5m). The fetch behind this — both a live plan/apply and collect-testdata — is chunked into day-sized requests regardless of how wide the range gets, so a large seasonal_

periods/window.lookback combination no longer risks exceeding a Prometheus-compatible backend's own per-series resolution limit (VictoriaMetrics/gigapipe's default 11,000 points) the way an unchunked request over the same range once could; it costs more Prometheus requests per run instead (one per day of range, per raw metric, per group) — a real cost worth sizing deliberately, not a reason to keep the range artificially small.

forecast.holt_winters.trend

One of add, mul, none; default none.

The trend component passed to the underlying Holt-Winters fit (statsmodels, an optional dependency — see 30-safety-and-status.md). The forecast runs a whole window.lookback ahead, and a trend is extrapolated undamped over all of it: on a real cluster add scored no better in the backtest than none but forecast a single disk at 95% of its whole group's load. Enable a trend only for load that really grows steadily, and check the forecast block's scaled disks afterwards.

forecast.holt_winters.seasonal

One of add, mul, none; default add.

The seasonal component passed to the same fit.

support — diagnostic bundles

See 26-collect-testdata-and-replay.md and IMPLEMENTATION_PLAN.md section 16. Nothing here is read on the normal plan/apply path — only by collect-testdata and by --replay's config loading.

support.salt_path

File path, default /var/lib/pve-storage-drs/anonymization-salt.

32 random bytes, generated on first use and persisted here, mode 0600, never written into a bundle. Every pseudonym collect-testdata produces is HMAC-SHA256 keyed on this salt, so the same object maps to the same pseudonym across runs and across bundles until --new-salt rotates it. Reproducibility is per salt file, i.e. per node by default; point this at a pmxcfs path (e.g. /etc/pve/pve-storage-drs-anon-salt) if every node in the cluster should agree on one mapping — that also replicates the salt to every node, which is the trade-off being made.

support.bundle_dir

Directory path, default /var/lib/pve-storage-drs/testdata.

Default -o/--output for collect-testdata when the flag is not given.

support.max_series_points

Positive integer, default 5000000.

collect-testdata prints the estimated series-sample count before fetching anything and refuses outright above this ceiling — never a silent truncation — naming the --range/--step/--no-series flags that would bring the estimate under it.

support.capture_range

The literal string "auto", or a duration, default "auto".

"auto" captures what holt_winters needs — currently max(2 * window.lookback, 2 * forecast.holt_winters.seasonal_periods * metrics.step) (twice the window, for the backtest) — so a bundle can reproduce a model the capturing operator never configured. An explicit duration (e.g. 14d) overrides that; --range on the command line overrides both.

`monitoring` — telling your monitoring system what the last run did

`monitoring.status_file`

String (a file path) or `null`, default `null`.

Where `apply` leaves its status report. `null` writes nothing. Set, the file is rewritten at the end of **every** `apply` run that produced a result — `dry-run` included, and runs that moved nothing included — in the format the `check_statusfile` plugin (Debian package `monitoring-plugins-contrib`) reads, so a plain `check_statusfile <path>` check is all a Nagios-style monitoring system needs. See `36-monitoring.md` for the format, what makes a run `OK`, `WARNING` or `CRITICAL`, and how to wire the check up.

Choose a path the user `apply` runs as can write and the monitoring user can read (the file is mode `0644`; the parent directory is created if it is missing). `/var/lib/pve-storage-drs/status` beside `state.json` is the natural place.

It interacts with the plugin's `--age`: the file's modification time is what says the timer is still firing, so a timer that runs every 15 minutes wants a check like `check_statusfile -a 1h`, while the plugin's default of 26 hours suits a daily one. Set too generous an age and a stopped timer goes unnoticed for that long; too tight and one slow run turns the check `WARNING`. A write failure is logged (`status_file_write_failed`) and never changes the run or its exit code.

Verifying your setup

Run this before relying on any plan. Telegraf and InfluxDB naming varies by deployment, so `pve-storage-drs` never assumes the defaults in `10-configuration.md` are correct for your cluster — `verify-metrics` is what checks them against the live Prometheus instead.

It also checks something the names alone cannot tell you: whether all six counters actually survived the transport. See `05-metrics-pipeline.md` for why they might not have, and why that produces a wrong plan rather than an error.

Running it

```
pve-storage-drs -c /etc/pve/drs.yaml verify-metrics
```

A successful run looks like this (labels and values are illustrative):

```
[   info] metrics.read_ops = 'blockstat_rd_operations' exists
[   info] metrics.write_ops = 'blockstat_wr_operations' exists
[   info] metrics.read_bytes = 'blockstat_rd_bytes' exists
[   info] metrics.write_bytes = 'blockstat_wr_bytes' exists
[   info] metrics.read_time_ns = 'blockstat_rd_total_time_ns' exists
[   info] metrics.write_time_ns = 'blockstat_wr_total_time_ns' exists
[   info] blockstat_rd_operations: sample series labels {'vmid': '101', 'instance': 'scsi0', 'nodename':
'pve01', 'cluster': 'pvezebe'}
...
[   info] 'cluster' label values seen across these metrics: pvezebe
...

OK: verify-metrics found no blocking problems
```

Exit status is `0` when every check passes, `1` when any check reports an `error`-level finding (see below). `--json` emits the same information as one JSON object instead — see `30-safety-and-status.md` for exit codes in general.

What each finding means

`verify-metrics` runs six checks, in this order (`IMPLEMENTATION_PLAN.md` section 3.3):

1. **Metric names exist.** Each of `metrics.read_ops`, `write_ops`, `read_bytes`, `write_bytes`, `read_time_ns` and `write_time_ns` is looked up against Prometheus's own `/api/v1/label/__name__/values`. A missing name is an **error** — fix the name in `10-configuration.md`'s `metrics.*` section to match what your Telegraf actually emits.
2. **Sample series and labels.** One instant query per metric, printing its full label set so you can see at a glance whether `metrics.labels.vmid`, `.device` and `.node` are really present and non-empty on a live series. A metric that exists but currently has no series is a **warning**, not an error — it may simply mean nothing has generated that kind of I/O recently. This step also cross-checks the full `(vmid, device)` set each of the six metrics reports against the other five — at no extra Prometheus cost, since it reuses these same six instant queries. If one metric is silently missing a disk the other five all report, that is a **warning** naming the metric and the disk: the usual cause is Telegraf's Prometheus-compatible output dropping a field the moment it sees a non-numeric value for it (InfluxDB's line protocol fixes a field's type from its first write, and Prometheus/OpenMetrics has no string sample type) — independently of the other five fields for the same disk, which is exactly why check 5 below (only looking at `read_ops`) cannot catch this on its own if a *different* field is the one affected.
3. **Configured labels present.** If `metrics.labels.vmid`/`.device`/`.node` do not appear (or are empty) on the sample series, that is an **error**: the load model has nothing to join disks on.
4. **The `instance` collision.** If `metrics.labels.device` is still the literal string `instance`, a **warning** — PVE's own per-disk tag has this name and it collides with Prometheus's unrelated scrape-target `instance` label. Many Telegraf configurations rename or overwrite one of the two; confirm which one survived before trusting this.
5. **Per-disk coverage.** For every `(vmid, device)` pair seen on `metrics.read_ops`, the fraction of expected samples actually present over `window.lookback` is measured. Below `window.min_coverage`, a **warning** naming the disk — that disk's own data will be rejected by the load model at plan time and its last known load from `state.json` used instead.
6. **Observed sample spacing.** The real gap between samples is measured (raw samples counted over a 10-minute window with `count_over_time()`, median across series), rather than trusted from `metrics.pvestatd_push_interval`. Disagreeing by more than 20%, or `metrics.rate_window` being below four times what was actually observed, is an **error** — this is what stops a stale `pvestatd_push_interval` declaration from silently breaking the `rate()` window it is supposed to protect.

If it fails

- **A metric does not exist**: check your Telegraf `influxdb_listener` (or equivalent) configuration for the actual measurement/field names it is producing, and update `metrics.*` to match.
- **A label is missing or empty**: the sample series printed by check 2 is the fastest way to see what labels Prometheus actually has for that series — update `metrics.labels.*` to name the real ones.
- **One metric is missing disks the others report**: a transport dropped a field. Check your Telegraf configuration (and the PVE-side field it collects) for that specific field emitting a non-numeric value for that disk at some point — that is enough to make Telegraf drop it permanently for that series, even while every other field for the same disk keeps working normally. This is a real defect in the pipeline, not noise, and it will not resolve itself; `05-metrics-pipeline.md` explains the mechanism, which output plugins are prone to it, and the gigapipe-on-ClickHouse setup that avoids the format bridge entirely.
- **Coverage is low for specific disks**: check that Telegraf is actually scraping every node, and that `window.lookback` is not longer than your Prometheus retention.
- **Spacing disagrees**: measure `pvestatd`'s real push interval on a node (its systemd timer, or the interval visible in `/etc/pve/status.cfg` if set there) and correct `metrics.pvestatd_push_interval`.

Reading `show-load` and `verify-storages`

Both commands only read — neither ever changes anything, in any `execution.mode`. Run them any time you want to see what the tool sees.

`show-load`

Prints every storage in every configured group, its disks, and its snapshot-reserve status. This example uses the group from `IMPLEMENTATION_PLAN.md` section 14's worked example (`config/drs.example.yaml` ships the same group and weights) so the numbers are traceable to that section rather than invented for this page:

```
$ pve-storage-drs -c /etc/pve/drs.yaml show-load
Group fc-tier1 → ACT: reserve violated on san-a; acting now regardless of the normal drift/imbalance
thresholds -- a capacity shortfall is never delayed by them
  san-a  provisioned 4.50 TiB/8.00 TiB  L=6.50 u=6.50  ⚠ reserve short by 2.50 TiB  (largest VM footprint
  3.00 TiB = web01(101), requires 6.00 TiB free)
    web01(101):scsi0             2.00 TiB  raw     ℓ 3.00
    web01(101):scsi1             1.00 TiB  raw     ℓ 1.00
    db01(102):scsi0              1.50 TiB  raw     ℓ 2.50
  san-b  provisioned 1.50 TiB/8.00 TiB  L=0.70 u=0.70  reserve OK  (largest VM footprint 1.00 TiB =
  app01(104), requires 2.00 TiB free)
    mail01(103):scsi0          512.00 GiB  raw     ℓ 0.40
    app01(104):scsi0             1.00 TiB  raw     ℓ 0.30
  san-c  provisioned 512.00 GiB/8.00 TiB  L=0.20 u=0.20  reserve OK  (largest VM footprint 512.00 GiB =
  cache01(105), requires 1.00 TiB free)
    cache01(105):scsi0         512.00 GiB  raw     ℓ 0.20
```

`largest VM footprint` is the quantity the snapshot reserve is computed from: a snapshot is taken of a whole VM, so what it needs on a storage is the sum of **all of that VM's disks on that storage** — here VM 101 holds a 2 TiB and a 1 TiB disk on san-a, a 3 TiB footprint, so san-a must keep `2.0 × 3.00 = 6.00 TiB` free, not the 4 TiB a rule looking only at the largest disk would ask for — and the VM named beside it is the one that drives the figure. Upgrading from a release that reserved only for the largest *disk*: a storage holding two or more disks of one VM can show a shortfall here that it did not have before. That is the rule being right, not the cluster changing, and `plan` treats it as a repair like any other — read this output first.

`provisioned` on a storage's line is the sum of every disk's *provisioned* size on it (the managed disks, plus any foreign volumes — templates, ISOs, backups, other groups' disks) against the storage's capacity — the figure the reserve shortfall beside it is computed from. The tool never counts on over-provisioning: a thin-provisioned pool (Ceph RBD, LVM-thin, ZFS) is counted at what its disks were provisioned at, not at what happens to be allocated. On a thick pool the two are the same number. On a thin one the pool's own, lower figure is printed next to it so the shortfall does not look like a mistake. A shortfall is counted in whole MiB, rounded up – a storage one byte short shows `⚠ reserve short by 1.00 MiB`, never `reserve OK`:

```
  ceph-a  provisioned 3.36 TiB/9.48 TiB (pool reports 1.45 TiB allocated)  L=2.10 u=2.10  reserve OK
  (largest VM footprint 1.00 TiB = app02(204), requires 2.00 TiB free)
```

Only the provisioned figure ever enters a decision; the allocated one is shown for information. (`--json` carries both: `provisioned_used_bytes` and `used_bytes`.)

`L=`/`u=` on a storage's line are `L_s` — the sum of `ℓ` over every disk currently on that storage — and `u_s = L_s / capability_weight`, the *utilization* that actually gets balanced across storages (so a storage configured with half the `capability_weight` of another is correctly expected to carry half the load, not the same absolute amount). `ℓ` after a disk's size/format is that disk's own load, measured in **average in-flight I/O requests**: by default it is `rate(rd_total_time_ns + wr_total_time_ns) / 1e9` for that disk — how

many read/write requests are, on average, in progress on the device at once — which self-weights by cost: an 8 ms write and a 1 ms read are correctly counted eight-to-one rather than as equal operations, the way a plain IOPS count would. (Read and write can be weighted asymmetrically via `load_weights.read_factor`/`write_factor`, and ops/bytes terms can be blended in alongside I/O time; see `IMPLEMENTATION_PLAN.md` section 4 for the full normalization and `docs/internals/70-loadmodel.md` for the code.) The worked numbers above are `IMPLEMENTATION_PLAN.md` section 14's own example (`config/drs.example.yaml` ships the same group and weights), not invented for this page.

Under `forecast.model: holt_winters` these are not the measured numbers. The default `quantile` model reports what was observed over the last `window.lookback`. With `holt_winters`, each disk's ℓ is scaled to a forecast of its p95 over the *next* `window.lookback`, and `L=`/`u=`, the spread and the `ACT`/`NO ACTION` verdict are all computed from those scaled values — they are the loads the balancer will act on. `show-load` says so on a line directly under the group header, the same line `explain` prints:

```
Group fc-tier1 → NO ACTION: imbalance 12.2% is below gates.imbalance_threshold (20.0%)
  forecast: holt_winters used (backtest error 0.08 vs baseline 0.14): 37 disks scaled to their forecast p95,
  5 kept as observed
```

or, when the forecast lost its backtest (or there is not yet enough history to run one) and the group fell back to observed loads, `forecast: holt_winters not used -- ...; loads are as observed`. No `forecast:` line at all means the `quantile` model is configured, or the forecast history could not be fetched this run (a warning is logged and the numbers are observed). See `explain` for the fields and `forecast.model` in the configuration reference for how the backtest decides.

The `Group <name> → ACT/NO ACTION` line

Four gates, evaluated in this order — **reserve override**, then the **capacity gate**, then **drift**, then **imbalance** — and the first one that decides wins. The header line above shows the reserve override: san-a's 0.5 TiB shortfall forces `ACT` outright, regardless of how balanced or drifted the group is, because "safety is not subject to hysteresis" (section 13). The other shapes this line can take, on a group with no reserve violation:

```
Group fc-tier1 → ACT: imbalance 255.4% meets or exceeds gates.imbalance_threshold (20.0%)
Group fc-tier1 → NO ACTION: imbalance 12.2% is below gates.imbalance_threshold (20.0%)
```

The **capacity gate** is the data-spread counterpart of the imbalance gate: instead of comparing I/O load across a group's storages, it compares how *full* each one is, as a fraction of its own capacity. It fires — bypassing drift and imbalance the same way the reserve override does, on the reasoning that a quiet workload is not a reason to leave data lopsided across storages — when the spread in fill fraction across the group's storages is at least `gates.capacity_spread_threshold` (default `0.25`) relative to the group's mean fill:

```
Group fc-tier1 → ACT: capacity spread 32.0% meets or exceeds gates.capacity_spread_threshold (25.0%) --
acting now regardless of I/O drift/imbalance
```

Unlike the other three, this gate can act on a group whose I/O is already perfectly balanced — `objective.delta_capacity_spread`, a price per TiB, is what does the actual spreading once the solver runs. Setting `gates.capacity_spread_threshold: null` disables it outright, and it never fires on a group with no data at all (mean fill `0`).

`show-load` reads `state.path`'s recorded load vector (section 11.2) and passes it to the drift gate, so once a group has one on record, this line can also read:

```
Group fc-tier1 → NO ACTION: drift 3.1% is below gates.drift_threshold (10.0%)
```

A group with no state.json, or none recorded for it yet, still evaluates as if this were the very first run — the drift gate is skipped outright (section 6's own degenerate-case rule), not "treated as zero drift" — so its verdict falls straight through to a reserve override, the capacity gate (which never depends on state.json at all), or an imbalance check, exactly as before this was wired up. docs/internals/80-gates.md and docs/internals/15-state.md have the detail; apply writes last_balance once a run actually executes at least one move (confirm or auto mode – dry-run only simulates, so it never triggers this), so this first-run behaviour is only the common case for a group before its first successful migration – after that, show-load's drift line reflects real history.

A pinned disk carries [pinned: <reason>] after its size and format — snapshots present (N), pending config change (unapplied), pending deletion (unapplied), locked: <lock>, excluded by config, excluded: unused disk (exclude.include_unused_disks=false), small disk with no larger disk of its VM in this group (migration.tiny_disk_bytes) (an EFI/TPM disk whose VM's other disks are all in a different group — it only ever moves with a larger disk of its VM), or cooldown: moved recently, <time> left on gates.cooldown_per_disk whenever state.json records that disk having moved within gates.cooldown_per_disk — matching IMPLEMENTATION_PLAN.md section 5.3 (C2) exactly; that reason is the answer to "why won't it move this disk." The two pending reasons (section 3.8) mean PVE has an edit or a deletion queued against that disk key that has not taken effect yet (usually waiting on a VM reboot) — apply or revert it in the PVE UI, then re-run.

A disk whose measured I/O falls below window.min_coverage (section 3.4) gets a ⚠ <disk> : <reason> line of its own, right after that group's disks, instead of a silently wrong ℓ — it either shows a state.json-recorded last known load (see docs/internals/70-loadmodel.md) or 0.0 when no state.json entry exists for it, and either way the warning says which. tpmstate0 and unusedN disks are never flagged this way: they genuinely emit no I/O metrics, so ℓ 0.00 for them is correct, not a gap. A group with no measured I/O at all gets one (idle: no measured I/O for this group this window) line instead of per-disk warnings — unless every one of the six queries came back with literally zero series (as opposed to some series with a genuinely-zero rate), in which case the line reads ⚠ the resolved query filter matched no series at all instead: that is a scoping problem (metrics.labels.node, or extra_selector), not an idle cluster, and verify-metrics's sample-series labels (docs/manual/20-verifying-metrics.md) are the next place to look — REVIEW.md W-06/W-07.

A Prometheus outage does not fail this command. Sizes and reserve status never depend on Prometheus at all; if the load fetch for a group fails, that group's storage/disk lines simply omit L=/u=/ℓ, and a ⚠ per-disk load unavailable: <reason> line explains why — check verify-metrics first if you see this.

A **Warnings:** block, when present, lists things worth a look but not fatal: a disk on a storage that is not in any configured group ("ungrouped, not managed" — check whether that storage belongs in a group), or a disk whose size came from its own config rather than the storage's authoritative content listing. The latter usually means the volume no longer exists on the storage — most often because it was removed directly on the storage backend rather than through Proxmox, which leaves a dangling reference behind that PVE does not clean up or even notice on its own (a VM starts fine with one; see IMPLEMENTATION_PLAN.md section 3.6 for a confirmed case). If you see this warning, check whether the named volume still exists on the storage and, if it genuinely does not, remove the stale reference (qm unlink <vmid> <device> for an unusedN entry).

--json emits the same information as one object with groups[].storages[] (including the exact byte counts behind the reserve check — largest_footprint_bytes/largest_footprint_vmid, the largest VM footprint on the storage and the VM it belongs to — plus load and utilization when available) and groups[].disks[] (plus load and load_flagged_reason when available), with each group carrying its own load_computed, idle, load_error and gate fields (plus a forecast object — model, used, backtest_error, baseline_error, disks_scaled, disks_kept — only under holt_winters, when the loads are forecast-

scaled) — a Prometheus outage on one group never prevents another group's `load_computed: true` or `gate` from being reported. `gate` is `null` when `load_computed` is `false` (no `GroupLoad` to evaluate gates against) and otherwise an object with `act` (bool), `reason` (string, identical to the human line's text after the arrow), `reserve_override` (bool), and `drift_fraction`/ `imbalance_fraction`/`capacity_fraction` (float or `null` — `null` means that gate was never reached, not that it evaluated to zero; `capacity_fraction` is the capacity gate's own ratio described above — `null` whenever `gates.capacity_spread_threshold` is unset or the group's mean fill is 0, the same two cases in which the gate itself never fires), and `split_vmid` (the id of the large VM that opened the group through the split gate — see `snapshot_reserve.split_vm_footprint` — or `null`).

A config with several groups issues Prometheus queries per group. Computing one group's load takes seven queries (six raw metrics plus one coverage check), each scoped to that group's own VMs — a config with `N` groups still makes `7N` queries in total on every `show-load`, not 7, since each group's coverage and data-quality decisions are independent and so cannot share a single fetch. For the typical one-to-three-group deployment this is a handful of fast instant queries and not worth worrying about; an operator running many groups against an already-busy Prometheus should be aware of the multiplier. Each of those queries only aggregates the VMs the calling group actually has (split into batches of `metrics.VMID_QUERY_BATCH_SIZE` VMs for a single very large group) rather than every VM in the cluster — on a cluster with hundreds of VMs, an earlier version of this tool aggregated the whole cluster on every one of those `7N` queries, which could make Prometheus itself time out the query outright rather than merely running slowly.

`verify-storages`

For a storage that sets `enforce_format` it also prints that value, the entry it came from, and how many disks already on the storage are in a different format (and their total size) — not an error, only how far the storage is from its policy, since enforcement applies to moves and never moves a disk by itself. The `--json` form carries the same as `enforce_format`, `enforce_format_source`, `nonconforming_disks` and `nonconforming_bytes`. It also carries `largest_footprint_bytes` and `largest_footprint_vmid` (the figure the snapshot reserve is computed from, and the VM it belongs to) beside `largest_disk_bytes` (the per-volume figure the wipe time is estimated from).

Reports the resolved `free_space.soft`/`.hard` bytes for each storage, each with the level it came from (`IMPLEMENTATION_PLAN.md` section 5.3.1) — the derivation an operator cannot otherwise predict, once inheritance, `/…/` patterns and percentages are all in play — plus the largest VM footprint on each storage with the VM it belongs to (what the snapshot reserve is computed from), `saferemove` and the wipe time it implies for the largest *disk* on each storage (a wipe is per volume), warning when your configured cooldown or move-duration limits are shorter than that implied wipe — the condition `IMPLEMENTATION_PLAN.md` section 9.3 describes as "the next run plans onto a storage that is still draining". Section 14's fixture has `saferemove` off everywhere (stated explicitly there, so its payback arithmetic is reproducible); this example instead assumes an LVM `san-b` with `saferemove` on at PVE's default 10 MiB/s throughput, to show what the warning looks like:

```
$ pve-storage-drs -c /etc/pve/drs.yaml verify-storages
Group fc-tier1
  san-a  saferemove=off
    largest VM footprint (Z_s): 3.00 TiB = web01(101)
    free_space: soft=0 B (global)  hard=0 B (= soft (no dip))
    saferemove is off or throughput unknown; no wipe-time check
  san-b  saferemove=on
    largest VM footprint (Z_s): 1.00 TiB = app01(104)
    free_space: soft=0 B (global)  hard=0 B (= soft (no dip))
    implied wipe time for the largest disk (1.00 TiB): 1.2d
    ⚠ gates.cooldown_per_storage (1.0h) is shorter than the implied wipe time -- the next run may plan onto
    a still-draining storage
```

```
    ⚠ migration.max_single_move_duration (6.0h) is shorter than the implied wipe time -- a move of the
    largest disk would be rejected outright
  san-c  saferemove=off
    largest VM footprint (Z_s): 512.00 GiB = cache01(105)
    free_space: soft=0 B (global)  hard=0 B (= soft (no dip))
    saferemove is off or throughput unknown; no wipe-time check
```

`free_space.soft`/`.hard` read `0 B` here because the fixture sets no `free_space` knob at all — the pre-section-5.3.1 default, unchanged. A config with a non-zero `free_space.soft` shows the resolved byte count here, the same value `plan`/`explain`/`show-load` all enforce.

The parenthesis after each value says where it came from, which is the part you cannot read off the config file when one line can mean a different number per LUN:

Shown	Meaning
`global`	inherited from the top-level `free_space` block
`storage entry`	written on this storage's own `groups[].storages[]` entry
`pattern /re/`	supplied by the `/…/` pattern entry that matched this storage
`…, 10% of 30.00 TiB`	appended when the value was a percentage: the capacity it was converted against
`= soft (no dip)`	(hard only) `hard` is null, so it equals the resolved soft

`--json` carries the same strings as `free_space_soft_source` and `free_space_hard_source` beside the two byte counts.

Storage types without `saferemove` at all (Ceph RBD, ZFS) always report `saferemove=off` and skip the check — there is nothing to wipe.

A negative `saferemove_throughput` is normal

If your `storage.cfg` reads

```
saferemove_throughput -1073741824
```

that is **1 GiB/s**, not an error and not a negative rate. PVE passes the value straight to `cstream -t`, where the sign chooses how the limit is enforced and the magnitude is the rate:

- **positive** — a session average. The wipe may exceed the rate for a while to make good on earlier underutilization, so that the run as a whole converges on it.
- **negative** — an upper limit on each individual read/write, never exceeded.

`verify-storages` reports the implied wipe time from the magnitude, so `-1073741824` and `1073741824` give the same duration; the `--json` output's `saferemove_throughput_bytes_per_sec` still echoes the value exactly as PVE has it, sign included, so you can compare it against `storage.cfg` without arithmetic. Everything that consumes the number — the migration cost the payback test charges, the `migration.max_single_move_duration` check, `apply`'s source-release wait — reads it the same way.

Versions up to 0.1.6 divided by the signed value instead, which produced a *negative* wipe time (`implied wipe time for the largest disk (1.00 TiB): -17.1m`). If you are on one of those, the two warnings above could never fire on a storage configured this way, and — where `|saferemove_throughput|` happens to equal `migration.bwlimit_bytes_per_sec`, a natural thing to configure — every move's total duration came out as exactly zero, so `migration.max_single_move_duration` was not enforced at all. Upgrading is the fix; no configuration change is needed.

If any `groups[].storages[].id` is written as a `/regex/` pattern (`IMPLEMENTATION_PLAN.md` section 11.4),

verify-storages also prints what each one matched this run, and lists any cluster storage matched by no group at all — the same "ungrouped, not managed" visibility show-load gives per disk, but at the storage level and independent of whether a disk currently happens to be on it:

```
Pattern expansions:
  [fc-tier1] /san-.*/ → san-a, san-b, san-c

Cluster storages matched by no group:
  - local-only
```

An over-broad or dead pattern is the one new way to misconfigure a group this feature adds, so this expansion is never silent: every run also logs it at INFO, whether or not verify-storages is the command being run.

Diagnostic bundles: collect-testdata and --replay

IMPLEMENTATION_PLAN.md section 16 is the specification; this page is the operational side of it. Two facts motivate the whole feature: one cluster is not a test corpus (every interesting bug so far was found by an operator running the tool on a cluster the author has never seen, then *describing* what they saw), and the data cannot leave the cluster as-is (a PVE inventory is customer names, hostnames, IQNs, ticket numbers and credentials).

collect-testdata: capturing a bundle

```
pve-storage-drs -c /etc/pve/drs.yaml collect-testdata
```

This is **read-only and dry-run-only** by construction: it never issues move_disk, and --mode confirm/--mode auto alongside it is a usage error (exit 2), not a silently ignored flag. It captures the *superset* of what any supported configuration could need — every raw metric over the history holt_winters needs (twice window.lookback, for its backtest), not just what the forecast.model you have configured needs — so a bundle captured today can still reproduce a Holt-Winters misfit tomorrow.

--estimate sizes the capture first: it reads the full PVE inventory (the same read pass a real capture or a plan run makes — every VM config, every content listing, every status call) to size the series capture, but fetches nothing from Prometheus:

```
pve-storage-drs -c /etc/pve/drs.yaml collect-testdata --estimate
```

```
groups: 2   disks: 47
range: 7d  step: 5m
estimated Prometheus queries: 117
estimated series sample points: 1137024
```

(The query count includes the day-sized chunking --estimate itself performs at capture time — 7 range-query chunks per group per metric at the 7 d default, not 1 — so it is the real HTTP request count, not a logical-query count that understates it.)

The sample-point figure is 47 disks · 6 metrics · 4032 points, not the 2016 a plain 604800s / 300s would suggest: at metrics.step == metrics.rate_window (both 5m here, and the project's own default) a live capture issues, and stores, range queries at *half* the configured step to work around a gigapipe backend bug (docs/internals/30-metrics.md), doubling the points actually captured. --estimate accounts for this so the printed figure — and the support.max_series_points refusal below, which compares against the same number — matches what a real capture stores, not the configured step's naive arithmetic.

Above support.max_series_points (default 5,000,000) the real capture refuses outright, naming the flags

that bring it under the limit:

Option	Effect
`-o, --output DIR`	Where the bundle directory and its `.tar.gz` are written (default `support.bundle_dir`).
`--estimate`	Read the full PVE inventory (one planning run's worth of API calls), print the estimate above, and exit; fetches nothing from Prometheus.
`--range DURATION`	Override the series capture range (default `support.capture_range`).
`--step DURATION`	Override the series resolution (default `metrics.step`).
`--no-series`	Topology, instant queries and findings only — no per-disk time series, so the bundle cannot exercise the `holt_winters` forecast, but is much smaller.
`--no-archive`	Write the directory only, skip the `.tar.gz`.
`--salt-file PATH`	Use a different anonymization salt (default `support.salt_path`).
`--new-salt`	Generate a fresh salt first. Logged at warning level: bundles made before and after no longer share a pseudonym mapping.

Exit status is `0` when every capture step succeeded, `1` when the bundle was still written but the manifest records one or more failed calls (a `Datastore.Allocate`-shaped permission gap looks exactly like this — see `00-installation.md`), `2` on a usage error.

What is in a bundle, and what is not

A bundle is a directory (`manifest.json`, `config.yaml`, `findings.json`, `pve/`, `prometheus/`, `SHA256SUMS`) plus a deterministic `.tar.gz` of it. **Read it before sending it anywhere** — it is plain, sorted, indented JSON, meant to be opened in an editor:

- **No direct identifier and no secret.** `config.yaml` never carries `proxmox.host/auth`, `prometheus.url`, or any other credential or endpoint — they are dropped, not blanked. Every node name, storage id, vmid, volume id, tag, pool and UPID is replaced by a pseudonym keyed to a salt (`support.salt_path`) that is generated locally, never written into the bundle, and never leaves the host. Free text — VM names, descriptions, snapshot names, storage comments — is dropped outright. Device keys (`scsi0`, `efidisk0`, ...) are not identifiers and are kept as-is, since the movability rules read them directly.
- `manifest.json` carries the **PVE and Prometheus server versions** (`capture.pve_version`/`capture.prometheus_version`, `null` if that one call failed) — not identifiers, so they need no anonymization, and they are the one piece of "what was this captured against" a bundle's own files otherwise leave to `tests/corpus/*.submission.yaml`, which only the committed corpus carries.
- **The mapping is stable per salt**, so a second bundle from the same host names the same objects the same way — useful for comparing a before/after pair — until `--new-salt` rotates it.
- **Sizes, capacities, load shapes and the whole configuration (minus credentials) are preserved.** That is what makes a bundle a test case instead of a shape. It also means a bundle is **not anonymous against someone who already knows the cluster** — a cluster with three nodes, two 8 TiB LUNs and 1,214 VMs is recognizable to anyone who has seen it. Submitting one to the project (see `tests/corpus/README.md`) is an informed decision, not a formality.

`--replay`: running against a bundle offline

```
pve-storage-drs --replay ./drs-testdata-cluster-3f8a91c2 plan --json
```

`--replay PATH` is a **global** option, accepted before any subcommand, and substitutes both the PVE and Prometheus clients at the one place each is constructed — so `verify-metrics`, `verify-storages`, `show-load`, `plan` and `explain` all run **unchanged**, with **no network access at all**. `PATH` is the **unpacked bundle directory**, not the `.tar.gz` it was sent as — `tar -xzf drs-testdata-cluster-3f8a91c2.tar.gz` first.

- The configuration is the bundle's own `config.yaml`, unless `-c` is also given — that is how you run the operator's cluster through a solver backend, a `spread_metric`, or a forecast model they never selected themselves.
- **`apply` is refused**, and so is any `--mode` above `dry-run` — exit `2` either way. A replay can never issue a write.
- A query the bundle has no recorded response for is a **loud, specific error** naming the query and the range it asked for — never a silent empty result standing in for "no data":

  ```
  pve-storage-drs: bundle has no recorded response for the range query
  'sum by (vmid, instance) (rate(blockstat_rd_operations{...}[300s]))'
  over [1700000000,1700086400] -- captured range was [1699400000,1700000000]
  ```

  This is the failure mode to expect if you hand-edit a bundle's `config.yaml` to a `window.lookback/metrics.step` combination the capture never anticipated, or if `--step` was overridden at capture time to something a replayed forecast does not expect.
- `state.json` is read from the bundle if present, and never written — a replay never touches the host's real state.

Run `--replay` against your own bundle before sending it anywhere: "does `plan` against this bundle show the problem I am reporting?" is exactly the question it answers, and it is also how the author verifies a submitted bundle reproduces what its `tests/corpus/*.submission.yaml` claims.

Sending one to the project

Bundles are wanted, and a pull request carrying one is as valuable as a pull request carrying code. A bug report describes a failure; a bundle reproduces it offline, on a machine that has never seen your cluster, for as long as the project exists. It is also the only thing that can prove the MILP solver and the fallback heuristic still agree at a few hundred disks — the synthetic fixtures in `tests/fixtures/` top out at six.

The terms, in short:

- **Read the bundle first.** Not the audit's word for it — your own. The scrub audit assumes the collector has a bug, which is the right assumption to make about a privacy control, but a clean audit is not consent.
- A bundle under **8 MiB unpacked** can be committed to the repository; larger ones stay outside it and are exercised from a directory named in `DRS_CORPUS_DIR`.
- It needs a `<bundle-name>.submission.yaml` recording who collected it, what it reproduces, that you reviewed it, and that you agree to it being redistributed under AGPL-3.0-or-later. Accepting one makes it public permanently, in git history.
- If your cluster is one whose size and shape alone are confidential, send a `--no-series` bundle, or just a description. A description is still worth having.

`tests/corpus/README.md` in the source tree (https://github.com/bzed/proxmox-storage-drs) is the full guide: the submission template, the four passes the suite runs over every committed bundle, and the consent terms in full.

Reading plan

plan computes what pve-storage-drs would do and prints it. It never changes anything, in any execution.mode — that is apply's job (see docs/manual/28-apply.md), and it needs a separate, explicit invocation to run at all.

For each group, plan:

1. Fetches the same load (ℓ_d/L_s/u_s) show-load does, and evaluates the same section 6 gate.
2. If the gate says NO ACTION, stops there for that group — no solver runs, nothing more to show.
3. If it says ACT, solves it with whichever backend solver.backend selects (see docs/internals/91-optimize.md for the CBC backend and docs/internals/90-heuristic.md for the dependency-free one) to compute a target assignment, orders its moves under the **transient reserve invariant** — while a move is in flight, the disk being migrated exists on *both* its source and target storage at once (the mirror is fully allocated on the target before the old copy is removed from the source), so the target's own snapshot reserve must already hold with that disk's bytes counted in, not only once the source frees them afterwards (IMPLEMENTATION_PLAN.md section 8.1 has the full formula, including the generalization to several moves landing on the same storage at once under concurrent execution — see docs/manual/28-apply.md) — and checks the whole plan against the payback rule below.

This example uses the same section 14 worked example show-load's manual page does, at the default weights (the two-move plan) and solver.backend: auto with neither MILP library installed, so it falls back to the heuristic — the numbers are traceable to that section either way, since every backend reports through the identical objective function:

```
$ pve-storage-drs -c /etc/pve/drs.yaml plan
Group fc-tier1 → ACT: reserve violated on san-a; acting now regardless of the normal drift/imbalance
thresholds -- a capacity shortfall is never delayed by them
  solver: heuristic
  1. db01(102):scsi0            san-a → san-c     1.50 TiB   ~2.2h   Δimbalance -4.53   ℓ/z 1.67
  2. web01(101):scsi1           san-a → san-b     1.00 TiB   ~1.5h   Δimbalance -2.00   ℓ/z 1.00
  after: san-a=3.00  san-b=1.70  san-c=2.70
  spread: 255.4% → 52.7%
  payback: benefit 1.93e+08 load·s vs cost 2.62e+04 load·s → ratio 7.34e+03 (need 10) ✓
```

The solver: line names whichever backend actually produced this plan – heuristic or cbc – plus, for the MILP backend, (optimal) or (feasible) (the gap was not proven closed within solver.time_limit_seconds, but a solution was still found). It is not always what solver.backend says: auto tries CBC before the heuristic, and even an explicitly forced cbc falls back to the heuristic (logged as a warning in that case) rather than failing the whole run, if that library is not installed or cannot produce a solution – see docs/internals/91-optimize.md. When CBC is what failed, the warning says why in CBC's own terms: hit solver.time_limit_seconds before finding a feasible solution means raising that limit may help; anything else (returned no usable solution (CBC status: Infeasible), for instance) is not a matter of time, and is worth a bug report with a collect-testdata bundle attached.

The after:/spread: lines (and the payback numbers) reflect what the scheduler actually managed to order, not the solver's target assignment — identical here since both moves scheduled cleanly, but see the deadlock note below for when a plan cannot schedule everything it proposed.

A group the gate does not act on prints one line and stops:

```
Group fc-tier2 → NO ACTION: imbalance 0.0% is below gates.imbalance_threshold (20.0%)
```

Reading a move line

`1. db01(102):scsi0 san-a → san-c 1.50 TiB ~2.2h Δimbalance -4.53 ℓ/z 1.67` — the disk (its VM's name, then its vmid in parentheses, then the device), its current and target storage, its size, the estimated mirror duration (`size / migration.bwlimit_bytes_per_sec`, plus a `+wipe <time>` suffix when `saferemove` on the source storage adds one — see the payback section below), this move's own effect on the group's imbalance metric at the moment it was scheduled (negative means imbalance went down, which is the usual case; a move scheduled mainly to resolve a reserve violation or consolidate a VM can show a positive value and still be correct), and `ℓ/z` — load per TiB, section 7.3's own "single best indicator of a good migration candidate: high I/O concentrated in a small disk."

A move whose own duration exceeds `migration.max_single_move_duration` carries a `⚠ exceeds migration.max_single_move_duration` suffix and always fails the plan (see below) — this is a hard, per-move rule, independent of whether the plan as a whole looks profitable.

A move scheduled first *regardless* of its ratio — because the storage it leaves is currently breaching its snapshot reserve or its configured `free_space.soft` requirement, the capacity a storage must always keep free, sized to the larger of `snapshot_reserve.factor` times its largest disk or `free_space.soft` (`IMPLEMENTATION_PLAN.md` section 5.3, constraints (C4)/(C5)/5.3.1) — is implied by the `ACT: reserve violated on ...` header in human output. Safety is not subject to hysteresis (section 13), so this always wins over a purely balance-driven move.

`repair` (visible in `--json`, `[repair]` in human output) marks a move the plan's own shortfall *depends on*: holding that one disk back on its current storage, in the plan actually executed, would leave the group short by more than it is. This is not the same question as "did this move's own source violate something" — an *indirect* repair (a move that empties the destination another repair needs) is marked too, even though its own source was clean, and a *redundant* repair (either of two moves alone would fix the violation) is marked on neither, since holding back just one still leaves the other to finish the job. See the `payback:` line below for what the plan as a whole being exempt actually depends on — a per-move `repair: true` marker is a report, not itself the trigger.

⚠ **`unfixable shortfall` means the plan cannot free what you asked for.** Every storage must end a plan with its snapshot reserve and its configured `free_space.soft` both free. When no placement of the disks this run is allowed to move gets there, `plan` (and `explain`, and `apply`) says so, with the amount still missing per storage, and lists the *pinned* disks sitting on those storages – the ones you can unpin (apply or revert a pending config change, clear snapshots, wait out a lock) to give the next run something to move:

```
Group tier2 → ACT: reserve violated on slow-a, slow-b; acting now regardless of the normal drift/imbalance
thresholds -- a capacity shortfall is never delayed by them
  solver: cbc (optimal)
  ⚠ unfixable shortfall: 231.05 GiB still short of the snapshot reserve / free-space requirement after this
  plan (slow-a 138.00 GiB, slow-b 93.05 GiB) -- no assignment of the movable disks can close it
    pinned on the short storages (unpin to let the next run move them):
      app01(144592):scsi0 on slow-a -- pending config change (unapplied)
      db02(337307):scsi0 on slow-b -- snapshots present (2)
```

A group whose gate forced `ACT` because of a shortfall and that then moves nothing is not being ignored: this line is the answer to "why not". Under `solver: cbc (optimal)` it is a proof – CBC first minimizes the total shortfall on its own, before anything else, so no placement of the movable disks does better. Under `solver: heuristic` the line says it is not proven. Shortfalls are counted in whole MiB, rounded up: one byte short is short, and prints as 1.00 MiB. The line repeats on every run until the cause is fixed, and the monitoring status file carries the same warning.

`no moves made` means the plan chose to stay put. The gate decides that a group is *planned*; whether

anything moves is the objective's call, and keeping every disk where it is can be the best plan. When the gate said ACT and nothing moves — and neither a deadlock nor an unfixable shortfall explains it — plan and apply say so:

```
Group tier1 → ACT: capacity spread 112.2% meets or exceeds gates.capacity_spread_threshold (25.0%) -- acting
now regardless of I/O drift/imbalance
  solver: cbc (optimal)
  no moves made: keeping the current placement scores best under the configured objective weights -- the
  gate decides that a group is planned, not that the plan must move something (explain shows the closest
  rejected move and why)
    the capacity gate opened this group: with beta 0.25 per move, gamma 0.05 and delta 0.5 per TiB only a
    disk larger than X* = 269 GiB can pay for its move; raise objective.delta_capacity_spread or lower
    objective.beta_move_count to spread data more aggressively, or raise gates.capacity_spread_threshold
    (null disables it) to stop planning for it
```

Under solver: cbc (optimal) "scores best" is proven; under the heuristic the line says it is not. The second line appears only when the capacity gate opened the group: there the I/O is typically balanced already, and the data-spread weight alone has to pay for every move. It is a price per TiB, so only a disk larger than the break-even size X* printed on the line can pay for its own move (the configuration chapter's objective.delta_capacity_spread entry derives it). After an imbalance gate the usual cause is a hotspot no placement can relieve, typically one disk carrying most of the group's load; explain's pinned-load line gives the best spread achievable. explain prints its own, fuller version: the closest rejected move and the objective arithmetic that rejected it.

A ⚠ line means a deadlock, not a hidden failure. If the target assignment includes a move this run cannot find any transient-feasible order for, plan says so explicitly (IMPLEMENTATION_PLAN.md section 8.3's "report, never force" path — staging and plan-splitting are not yet implemented, so this is where a genuine cycle or an unavoidable capacity shortfall currently surfaces) rather than silently omitting the move or, worse, telling you a plan is clean when part of it is not achievable. When only *some* moves deadlock, the after:/spread:/payback: lines report the state reachable by the moves that did schedule, never the fuller picture the undeliverable ones would have produced.

The split: line

When a group holds a VM larger than snapshot_reserve.split_vm_footprint (default 2 TiB), the group's output carries one line per such VM that the plan changes, or that is still above its cap afterwards:

```
  split: VM 701 largest footprint 10.00 TiB → 4.00 TiB (cap 2.00 TiB, excess 8.00 TiB → 2.00 TiB — the
  solver found no better split across 3 storages)
```

Largest footprint is the most of that VM any one storage holds (the sum of its disks there — the figure a snapshot of the VM has to find room for, doubled at the default reserve factor). *Cap* is the threshold the VM is spread towards (a single disk larger than the threshold raises its own VM's cap to that disk's size). *Excess* is how far the largest footprint is above the cap. With fewer storages than disks the cap is usually unreachable, and the line says so: a result still above the cap ends "the solver found no better split across 3 storages" when the MILP settled on it (the solver minimizes the whole objective within solver.mip_gap, so this is not a proof that no other split has a lower peak), and "still above the cap" otherwise. A group is planned for this reason even when its I/O and data are in balance (the gate reads "VM 701 holds … against a … cap … acting now regardless of I/O drift/imbalance"), and such a plan has to pay for itself like any balance plan — splitting is a preference, not a repair, and gets no exemption from the payback: test below.

The `payback:` line

Section 7.3's acceptance test: `benefit >= migration.payback_ratio * cost`, both sides in load-seconds. `benefit` is `(alpha_spread * ΔE + delta_capacity_spread * ΔF + kappa_vm_affinity * ΔA + mu_vm_split_per_tib * ΔO) * migration.payback_horizon` (O is the summed split excess of large VMs, see "The `split:` line" below; section 7.2's own *unweighted* E/F/A/O, not the solver's already-scaled objective terms — passing those instead would double-apply the weight); `ΔA` may be negative, and then it reduces the benefit — a balance move that splits a VM pays for that fragmentation out of its other gains. `E_after`/`F_after`/`A_after` are evaluated against what `plan`'s own scheduler actually managed to order, not the solver's aspirational target, so a partially-deadlocked plan is scored on the moves it can really make, not ones it cannot. `cost` sums every *scheduled* move's `duration_mirror * (source_load_weight + target_load_weight) + duration_wipe * wipe_load_weight`, except a disk below `migration.tiny_disk_bytes`, which costs `0` regardless of duration (section 7.1). A ✓ plan passed; a ✗ one did not. A plan made entirely of moves below `migration.tiny_disk_bytes` has `cost = 0` and renders as `ratio inf (need 10) ✓` — a real, correct verdict, not a formatting bug.

A ✗ plan can fail for two different reasons, reported as two different lines, because they call for different fixes:

- **Economic failure** — the balance benefit does not outweigh the migration cost. Adjust `objective`'s weights, `migration.payback_ratio`, or accept the plan is not worth running.
- **Hard-duration failure** — some move's own `duration_mirror + duration_wipe` exceeds `migration.max_single_move_duration`, regardless of whether the plan as a whole is profitable. Fix `saferemove` throughput, `migration.bwlimit_bytes_per_sec`, or the duration limit itself.

Either, both, or neither can apply to the same plan; `plan` names exactly which happened rather than one generic "does not pass the payback test" line for both.

A plan that leaves the group with less reserve/free-space shortfall than it found always passes this test, regardless of the ratio shown — the example above happens to pass on merit (133 ≥ 10), but a plan whose *only* move fixes a snapshot-reserve or `free_space.soft` violation with zero balance benefit (a real, common case: relocating the sole loaded disk in a two-storage group changes which side carries it without reducing spread at all) is accepted too. This is `--json`'s `payback.repair_exempt`, decided on the plan's *outcome* — its executed endpoint's `Σ r_s` strictly below the current assignment's (`payback.reserve_shortfall_bytes_before`/`_after`) — not on any one move's own flag, so a plan that moves a disk off a violating storage but leaves the group no less short in total is **not** exempt. Section 13's "the reserve is never traded against balance" applies here exactly as it does to the gates: an operator does not get to decline a capacity emergency fix because it scores poorly against `migration.payback_ratio`. The hard per-move duration rule is not exempted this way — it is an operational limit, not an economic one, and still blocks the plan, and a move it excludes is also excluded from what `repair_exempt` scores: a repair a hard rule then blocks is correctly not exempt either.

What a ✗ (or a rejected move) does *not* do today: automatically make the plan smaller and retry. Section 7.3 describes re-solving with `beta` and `gamma` doubled, up to three times, converging on the highest-value subset of moves. That loop is not implemented — a failing plan is reported and left for you to review, not silently adjusted. See `docs/internals/96-payback.md`.

What `plan` does not yet do

- **No pinned block, fragmentation naming, or pinned-load line.** Unlike `IMPLEMENTATION_PLAN.md` section 9.5's own example, `plan`/`apply` print only the move list and payback verdict — that narration (including the per-pin → action hint) lives in `pve-storage-drs explain` instead (`docs/manual/29-explain.md`); `show-load` also shows each pin inline, per disk.

- **No automatic re-solve-and-shrink on a failing payback test** (see above) — reported, not fixed for you.
- **`plan` itself still only reads `state.json`, never writes it.** Its gate reads real `last_balance` history when a group has one recorded (see `docs/internals/15-state.md`), and a disk/storage cooldown pins or excludes exactly as described below – but `plan` never executes a migration, so it never has anything of its own to record. `apply` is what writes `last_balance` and cooldowns now, after a run that actually executed at least one migration (`docs/manual/28-apply.md`); a group with no `state.json` yet, or one `apply` has never touched, still evaluates as a first run (drift gate skipped).
- **Cooldowns.** A disk moved within `gates.cooldown_per_disk` is pinned and excluded from this run's solve — `plan` itself does not print pins at all (see above); `show-load`/`explain` both show it as `[pinned: cooldown: ...]`. A storage that was a migration's *destination* within `gates.cooldown_per_storage` accepts no new incoming moves from the heuristic. See `docs/internals/15-state.md`, `docs/internals/60-topology.md` and `docs/internals/90-heuristic.md`.
- **No staging, no concurrent scheduling.** Documented as deliberate, not forgotten, in `docs/internals/95-schedule.md`.

`--json` emits `groups[]`, each with `gate` (identical shape to `show-load`'s), `solver_backend`/`solver_status` (`null`/`null` when the gate said `NO ACTION`; otherwise `"heuristic"`/`null`, or `"cbc"` with `"optimal"`/`"feasible"` – the same information the human output's `solver:` line names), `moves[]` (`disk_key`, `vmid`, `vm_name`, `device`, `from_storage`, `to_storage`, `size_bytes`, `imbalance_reduction`, `format_from`/`format_to` (the disk's format before the move and the one it arrives in — equal unless the target storage sets `enforce_format`; `null` on a hand-built move), `repair` (the section 7.3 revert-test marker — see "The `payback:` line" above), `load_per_tib`, `duration_mirror_seconds`, `duration_wipe_seconds`, `cost_load_seconds`, `exceeds_max_duration`), `vm_splits[]` (one entry per `split:` line: `vmid`, `cap_bytes`, `peak_footprint_bytes_before`/`_after`, `excess_bytes_before`/`_after`), `deadlocked` (a list of disk keys) and `deadlock_message` (`null` if none), `before_spread`/`after_spread` (the section 6 spread fraction, before the plan and after every scheduled move), `before_capacity_spread`/ `after_capacity_spread` (the *relative* fill-fraction spread the capacity gate compares, `(max − min) / mean` of bytes used / capacity, a fraction and not the TiB term `objective.delta_capacity_spread` prices, the same way, `null` for a group whose mean fill is 0 or that has not solved), `before_objective_total`/`after_objective_total` (the section 5.4 objective's full `.total`, `evaluate_assignment()` re-scored at the same true weights `validate_corpus.py`'s cross-backend check uses – REVIEW.md AA-01), `load_error` (`null` unless Prometheus failed for this group), and `payback` — `null` when there is no `GroupLoad` or the gate said `NO ACTION`, otherwise an object with `benefit_load_seconds`, `total_cost_load_seconds`, `ratio`, `aggregate_ok` (the economic test alone, or `true` if exempted), `repair_exempt` (the outcome trigger itself — see above), `reserve_shortfall_bytes_before`/ `_after` (Σ `r_s` on the current assignment and on the plan's executed endpoint, what `repair_exempt` is decided from), `split_excess_bytes_before`/ `_after` (the summed split excess of the group's large VMs, in bytes, the same pair of figures the `split:` lines are drawn from), `rejected_moves` (disk keys failing the hard duration rule) and `accepted` (`aggregate_ok` and `rejected_moves` empty). `unfixable_shortfall` is `null` unless the plan leaves a storage short, and then an object with `total_bytes`, `by_storage` (storage id to bytes still short, whole MiB rounded up), `proven` (`true` only for `solver_backend: "cbc"` with `solver_status: "optimal"`) and `pinned_blockers` (`disk_key`, `storage`, `reason` for every pinned disk on a short storage). Under `forecast.model: holt_winters` the group also has a `forecast` object: `model`, `used`, `backtest_error`, `baseline_error`, `disks_scaled`, `disks_kept` (absent under the default `quantile`).

Reading `apply`

`apply` runs the identical per-group pipeline `plan` prints — the same gate, the same solver, the same scheduler, the same payback test (`plan`'s own manual page, `docs/manual/27-plan.md`, describes all four

in detail) — and then, unless it is dry-run, actually issues the moves. execution.mode (or --mode) picks what happens next:

Mode	What apply does
dry-run *(default)*	Prints the same report plan would, plus a would_move outcome per move. Issues zero API calls.
confirm	Executes one move at a time, prompting before each: [y]es/[n]o skip/[a]ll remaining/[q]uit.
auto	Executes unattended, subject to execution.time_windows, max_migrations_per_run, and a bounded automatic re-plan loop — see "Reading auto mode" below. Also the only mode that honours execution.max_concurrent_migrations/max_concurrent_per_storage above their default of 1, running several moves at once — see "Concurrent execution" below.

A confirm run, using the same section 14 fixture plan's own manual page walks through, with the operator declining the first move and accepting the rest:

```
$ pve-storage-drs -c /etc/pve/drs.yaml --mode confirm apply
Group fc-tier1 → ACT: reserve violated on san-a; acting now regardless of the normal drift/imbalance
thresholds -- a capacity shortfall is never delayed by them
  solver: heuristic
  db01(102):scsi0  san-a → san-c  1.50 TiB  [y]es/[n]o skip/[a]ll remaining/[q]uit? n
  1. db01(102):scsi0            san-a → san-c     1.50 TiB   ~2.2h   Δimbalance -4.53   ℓ/z 1.67  → skipped:
  operator declined
  web01(101):scsi1  san-a → san-b  1.00 TiB  [y]es/[n]o skip/[a]ll remaining/[q]uit? a
  2. web01(101):scsi1           san-a → san-b     1.00 TiB   ~1.5h   Δimbalance -2.00   ℓ/z 1.00  → moved:
  task UPID:... completed OK, source released
  after: san-a=5.50  san-b=1.70  san-c=0.20
  spread: 255.4% → 214.9%
  payback: benefit 3.02e+07 load·s vs cost 1.05e+04 load·s → ratio 2.88e+03 (need 10) ✓
```

Everything above the → status: detail suffix on each move line is identical to plan's own output — the "after"/"spread"/"payback" lines still describe the plan as scheduled, not adjusted for what a declined move would have changed, since (as with a partial deadlock, see plan's manual page) reporting what was actually asked for is more honest than recomputing a hypothetical.

The [y]es/[n]o skip/[a]ll remaining/[q]uit prompt

Only in confirm mode, once per move, before it is issued — never for a move dry-run would only report. [a]ll remaining stops prompting for the rest of *this run* (every group, not just the current one); [q]uit stops the whole run immediately, including any group not yet even planned. An answer that is not one of the four is rejected right there ("please answer y, n, a or q") and asked again — it never reaches execute.py at all, so a typo can never be mistaken for one of the four real decisions.

Reading auto mode

auto behaves like [a]ll remaining chosen up front, unprompted, plus three things dry-run/confirm do not do at all (IMPLEMENTATION_PLAN.md section 9.1/9.2):

- **execution.time_windows.** If any are configured, a move is refused (reported skipped, "outside execution.time_windows") unless the current moment — in the **local time of the host running pve-storage-drs**, not UTC — falls inside one of them, and before starting a move auto also refuses it if its estimated duration (mirror plus any saferemove wipe) would not finish before the window closes. No windows configured at all means no restriction. Either way, a move already in progress is never aborted at window close — the check only ever runs *before* issuing the next one.

- **`execution.max_migrations_per_run`.** A ceiling on how many moves the whole invocation executes, shared across every group it visits (not reset per group) — the remainder waits for the next scheduled run.
- **The re-plan loop.** A `replan_needed` mismatch does not end an `auto` run the way it ends a `dry-run`/`confirm` one: `apply` re-invokes the whole pipeline (gates, load model, solver, payback, ordering) from freshly observed cluster state and tries again, up to `execution.max_replans_per_run` times. The gate may well conclude no further action is needed on the re-plan — a normal, quiet outcome, not a failure. Exceeding the cap ends the run with a message naming the setting, and the run still exits `0`: too many outside changes (VMs moved, disks created, another tool filling a target) to plan against is a reason to try again later, not a failure of the tool, and the next run starts from freshly observed state. A re-plan that cannot be computed because Prometheus cannot be read is different — see “When something cannot be read” below. Cross-referencing the report's `outcomes[]` shows the whole story for a re-planned group: the mismatch that triggered each re-plan, and whatever the next attempt then did.

Concurrent execution

Configuring `execution.max_concurrent_migrations`/`max_concurrent_per_storage` above their default of `1` runs several moves at once in `auto` mode (`dry-run`/`confirm` always run strictly sequentially, regardless of these settings — concurrency only makes sense for unattended operation). The **transient reserve invariant** — the rule (`docs/manual/27-plan.md`) that a storage's snapshot reserve must hold even while a disk migrating onto it exists on both source and target at once, not only once the move finishes — generalizes to a whole in-flight set of moves landing on the same storage at the same time (`IMPLEMENTATION_PLAN.md` section 8.1's own formula for it), checked live before every launch exactly like the sequential executor's own pre-flight re-check; `max_concurrent_per_storage` counts a storage as occupied whether a move touches it as source *or* target.

Launch order stays strictly FIFO. The scheduler's own queue (the same one a sequential run would follow, one move at a time) is never reordered to keep every concurrency slot busy: if the next queued move cannot launch yet — its VM is locked, or a per-storage cap is already reached — `apply` waits for it rather than skipping ahead to a later move that could launch immediately. Several moves genuinely run at once once launched (nothing blocks waiting for one move's own completion or lock to affect any other), but which move launches *next* is always decided in the order `plan` would have shown it. This can under-deliver on throughput in a mixed queue, but never launches a move out of the order the payback-scored plan itself computed.

A failure (or a budget/deadline limit) stops the run from launching anything further, exactly as in the sequential case, but does not abandon moves already in flight: `apply` keeps polling them to their own natural conclusion (`moved`, `failed`, or `draining`) before returning, so nothing already running is left unreported.

What “done” means for one move, and why it can take a while

A move is not finished when the `move_disk` task itself succeeds. `IMPLEMENTATION_PLAN.md` section 9.3 requires all three, together: the task reports `exitstatus: OK`, the source volume is gone from the source storage's own content listing, and the VM's config lock is clear again. A storage with `saferemove` enabled zeroes the old volume afterwards at `saferemove_throughput` (default 10 MiB/s; a negative value in `storage.cfg` is a rate too, see `verify-storages`) — for a large disk, that can run for hours after the task itself is long done. While it runs, `apply` reports the move as `draining`, not `moved` and not `failed`: it is a normal, expected state, bounded by `execution.source_release.timeout` (default 48h), not an error. A `draining` move still counts as “executed” for `state.json`'s bookkeeping below — the mirror itself completed; only the source's own cleanup is still in flight. For the *rest of this same run*, that storage is

excluded as both a source and a target for any later move: it holds a storage-level lock for as long as the wipe runs, so a subsequent `move_disk` touching it would either queue behind a wipe that can take days or fail outright. A move skipped this way is reported plainly (`skipped`, naming the draining storage) rather than attempted into that lock.

Before *every* move, `apply` re-checks the live cluster rather than trusting the plan: the VM may have moved node, the disk may no longer be on the expected source, a snapshot may have appeared, it may have been tagged for exclusion, or the target storage's free space may no longer satisfy the transient invariant (the target must still hold every disk on it, plus the disk being moved, plus the snapshot reserve, while both copies of the moving disk exist). That last re-check reads the target's volume listing afresh and counts every volume at its *provisioned* size, exactly as planning did — not the pool's own allocated figure, which on a thin-provisioned pool is much lower and would let a move through onto a pool that other provisioning has since filled. A mismatch — the VM moved, a snapshot appeared, the target no longer has room — stops the group's run with `replan_needed`. In `dry-run`/`confirm`, that is the end of it — the operator re-runs `apply` by hand once ready; see "Reading `auto` mode" below for what `auto` itself does about it automatically.

If a check cannot be *made* at all — the PVE API errors while re-reading the VM or the target, or the target lists a volume with no size — the move is refused and the whole run fails; see "When something cannot be read".

A VM config lock (`backup`, `snapshot`, `migrate`, or any other value — this set is never whitelisted, see `docs/internals/92-execute.md`) makes `apply` wait, not fail, up to `execution.locks.wait_timeout_seconds`. So does a task still running on the VM — typically the "Erase data" job (`imgdel`) of an earlier move — because it can hold the config file's own lock and would make the move fail with `can't lock file ... - got timeout`; console sessions do not count. It is the same wait, with the same timeout and `on_timeout` handling. What happens on that timeout depends on `execution.locks.on_timeout`: `skip` (the default) reports the move `skipped` and continues with the rest of the plan; `abort` reports it `failed` and stops the *whole run* unconditionally — this one case ignores `execution.abort_on_failure` entirely, because the manual's own words for `on_timeout: abort` are "abort the run," a distinct, stronger promise than the general "stop after any failed move" policy a plain task failure still goes through.

Failure and `abort_on_failure`

A failed `move_disk` task is reported `failed` with its detail; `apply` then lists any volume left behind on the target that is not referenced in the VM's config (`IMPLEMENTATION_PLAN.md` section 9.4) — **never deleted, only reported**, exactly like every other orphan this project surfaces. With `execution.abort_on_failure: true` (the default), the whole run stops there; with it `false`, `apply` continues with the plan's remaining moves. The lock-timeout-`abort` case above is the one exception that always stops the run regardless of this setting.

When something cannot be read

A mismatch between the plan and the cluster is something `apply` can plan around; a *failure to look* is not. These end the whole run at once — no later group is planned or executed — and `apply` exits `1`:

- the PVE API errors while `apply` re-reads a VM or the target storage just before a move (or the target lists a volume with no size);
- Prometheus cannot be read while computing a group's load — in the first plan, or in an `auto` re-plan. (A re-plan that cannot be computed is a failed run, not "no further action needed".)

Moves that already completed stay recorded in `state.json` and in the report, and the report says what could not be read. Planning around data it could not observe would be a worse plan, not a better one, so

the tool does not try: fix the connection and run it again. `plan` and `explain`, which change nothing, still report every group they can — and also exit `1` when any group's load could not be computed.

`state.json`: what a real run actually changes

`apply` is the one command that takes `state.json`'s advisory lock (`docs/internals/15-state.md`) — checked first, before it ever touches PVE or Prometheus, so a second concurrent `apply` (any mode, including `dry-run`) exits `0` immediately and quietly rather than racing the first one. After the run, for every group where at least one move actually executed (`moved` or `draining` — both mean the mirror itself completed):

- `last_balance.load_vector` is replaced with that group's load for this run — the load the balancer acted on, so forecast-scaled under `forecast.model: holt_winters` — so the next run's drift gate compares against a real balance instead of treating every run as the first one.
- Every executed disk gets a fresh `gates.cooldown_per_disk` timestamp at its **new** location.
- **Both** of the move's storages — source and destination — get a fresh `gates.cooldown_per_storage` timestamp: section 6 says a storage "involved in a migration" accepts no new incoming moves, and section 9.3's sizing rule for this cooldown is specifically about protecting a *source* still draining a `saferemove` wipe, so recording the destination alone could never deliver on it. What the cooldown actually *excludes* on the next run stays destination-only (the heuristic/MILP backends refuse a cooldown storage as an incoming-move target, never as a source a disk may still leave) — recording both endpoints only widens which storages carry a timestamp, not what that timestamp blocks.

A group that never executes anything (the gate said `NO ACTION`, or a `load_error`/`replan_needed` stopped it before any move ran) leaves `state.json` untouched for that group.

Crash and two-instance recovery

Before planning anything, every `apply` run (including `dry-run`) checks for a `move_disk` left running by a crashed previous run or a second, concurrently-running instance: `state.json`'s own `inflight_upids` are re-checked, and every node's task list is scanned for a still-running `qmmove` task from this tool's own configured user. Any vmid this finds is excluded from this run's planning exactly like a manually-configured `exclude.vmids` entry — it shows up pinned in the report with the reason "excluded by config (`exclude.vmids`)", and a log line at `WARNING` explains which check actually found it. It is never force-cancelled, and this run never assumes it is safe to touch that VM's disk again.

During execution, `state.json`'s `inflight_upids` is written **before** each `move_disk` is issued and cleared once that task itself has finished — synchronously, so a crash (killed, OOM, host reboot) partway through a move leaves a trace the *next* run's own startup check will find, rather than only ever being updated at the very end of a whole `apply` run. See `docs/internals/93-crashrecovery.md` for the full mechanism.

`--json`

Identical to `plan`'s own `groups[]` shape (see `docs/manual/27-plan.md`), with one addition per group: `"execution"`, either `null` (never reached — a `load_error`, `NO ACTION`, or a group the run never got to because an earlier one stopped it, e.g. an operator's `[q]uit`) or an object with `stopped_early`, `stop_reason` (`null` unless the run stopped for this group specifically), `aborted` (`true` when this group's execution hit one of the failures above and the run ended because of it), `replans_exhausted` (`true` when `execution.max_replans_per_run` ran out — the run bailed out and exits `0`) and `outcomes[]` — one entry per move actually attempted, each with `disk_key`, `vm_name`, `from_storage`, `to_storage`, `status` (`would_move`/`moved`/`skipped`/`failed`/`draining`/`replan_needed`), `detail`, `upid` (`null` unless `move_disk` was actually issued for this move — always `null` for `would_move`/`replan_needed`, and for `skipped` except a lock timeout that happened after issuing nothing yet, so still `null` there too; a `failed` outcome carries one

only when the task itself ran and failed, not when a lock timeout aborted before it started) and orphaned_volumes (only ever non-empty after a failed outcome).

Reading explain

explain runs the identical gate/solve/schedule/payback pipeline plan does (docs/manual/27-plan.md) — same load, same gate, same solver, same scheduler, same payback test, byte-for-byte the same numbers — and prints everything plan already shows, plus the "why" plan itself never does: the measured load every one of those numbers derives from, which disks are pinned and the exact reason, which VMs a pin leaves spread across more than one storage, the section 5.4 objective broken into its seven terms, and whether the pinned load is large enough that the residual imbalance is structural rather than a planning shortfall. -v additionally names exactly which Prometheus query produced all of it. It never changes anything, in any execution.mode — exactly like plan.

This example extends the section 14 worked example plan's own manual page uses with one more VM, 106 (archive01): its scsi0 carries two real snapshots, so it is pinned (section 3.7), while its scsi1 is an ordinary movable disk on a different storage — the "cannot fully consolidate" case section 3.6 exists to name:

```
$ pve-storage-drs -c /etc/pve/drs.yaml explain
Group fc-tier1 → ACT: imbalance 255% exceeds gates.imbalance_threshold (20%)
  solver: heuristic
  1. db01(102):scsi0            san-a → san-b     1.50 TiB   ~2.2h   Δimbalance -4.00   ℓ/z 1.67
  2. web01(101):scsi0           san-a → san-c     2.00 TiB   ~2.9h   Δimbalance -3.40   ℓ/z 1.50
  3. mail01(103):scsi0          san-b → san-a   512.00 GiB   ~43.7m   Δimbalance -0.80   ℓ/z 0.80
  4. cache01(105):scsi0         san-c → san-a   512.00 GiB   ~43.7m   Δimbalance -0.40   ℓ/z 0.40
  after: san-a=2.50  san-b=2.90  san-c=3.00
  spread: 257.1% → 17.9%
  payback: benefit 2.52e+08 load·s vs cost 4.72e+04 load·s → ratio 5.34e+03 (need 10) ✓
  objective: imbalance 0.6 + moves 1 + bytes 0.225 + fragmentation 1.43 + spread 0.25 + split 0 + reserve 0
  = 3.5
  measured load:
  san-a  provisioned 5.50 TiB/8.00 TiB  L=7.40 u=7.40  ⚠ reserve short by 3.50 TiB  (largest VM footprint
  3.00 TiB = web01(101), requires 6.00 TiB free)
    web01(101):scsi0             2.00 TiB  raw     ℓ 3.00
    web01(101):scsi1             1.00 TiB  raw     ℓ 1.00
    db01(102):scsi0              1.50 TiB  raw     ℓ 2.50
    archive01(106):scsi0         1.00 TiB  raw     ℓ 0.90  [pinned: snapshots present (2)]
  san-b  provisioned 2.00 TiB/8.00 TiB  L=0.80 u=0.80  reserve OK  (largest VM footprint 1.00 TiB =
  app01(104), requires 2.00 TiB free)
    mail01(103):scsi0          512.00 GiB  raw     ℓ 0.40
    app01(104):scsi0             1.00 TiB  raw     ℓ 0.30
    archive01(106):scsi1       512.00 GiB  raw     ℓ 0.10
  san-c  provisioned 512.00 GiB/8.00 TiB  L=0.20 u=0.20  reserve OK  (largest VM footprint 512.00 GiB =
  cache01(105), requires 1.00 TiB free)
    cache01(105):scsi0         512.00 GiB  raw     ℓ 0.20
  pinned (not movable this run):
    archive01(106):scsi0         1.00 TiB  on san-a  ℓ 0.90  ℓ/z 0.90  -- snapshots present (2)  → clear
    snapshots to unblock
  cannot fully consolidate:
    106 (archive01)  scsi0: snapshots present (2)
  pinned load 0.90 of 8.40 (10.7%, warn at 25%);  best achievable spread given pins: 17.9%
```

Everything up to and including the payback: line is identical to plan's own output for the same input — see docs/manual/27-plan.md for how to read the move lines and the payback verdict. What follows is new.

The `forecast:` line

Printed only under `forecast.model: holt_winters`, just before the `objective:` line, once per group whose load was computed:

```
forecast: holt_winters used (backtest error 0.08 vs baseline 0.14): 37 disks scaled to their forecast p95,
5 kept as observed
forecast: holt_winters not used -- it did not beat the quantile baseline on this group's recent history
(or there is not enough of it yet); loads are as observed
```

`used` means the group's per-disk loads (the ℓ in every number above and below) are scaled to a Holt-Winters forecast of their p95 over the next `window.lookback`; the two errors are the backtest that decided it (see `forecast.model` in `docs/manual/10-configuration.md`). "Kept" disks are flagged for low coverage, idle, or had no fit. `plan --json` and `explain --json` carry the same facts as a per-group `forecast` object, and `show-load` prints the same line (and `--json` object) so its loads are never mistaken for observed ones.

The `objective:` line

The section 5.4 objective the solver actually minimized, broken into its seven terms rather than only the total — `imbalance` (`alpha` times the spread metric), `moves` (`beta` times the move count), `bytes` (`gamma` times TiB moved), `fragmentation` (`kappa` times each VM's I/O-weighted extra storage count beyond one — section 5.4's `w_v`, so a VM doing several times the group's average I/O costs correspondingly more to split), `spread` (`delta` times the section 5.3 (C7) data-spread deviation), `split` (`mu` times the TiB by which a VM above `snapshot_reserve.split_vm_footprint` exceeds its cap on its fullest storage — zero when no VM is that large), and `reserve` (the penalty for any remaining (C5) shortfall, zero on a plan that resolves or never had one). This is the same `heuristic.ObjectiveBreakdown` the solver itself compares candidate assignments with — the reason that class keeps the seven terms apart instead of collapsing to only `.total` in the first place. Only printed when the gate said `ACT`; a `NO ACTION` group solved nothing this run, so there is no objective to show.

`no moves made: ... / closest alternative: ...`

Only printed when the gate said `ACT` *and* the solver's own optimum still moves nothing — every disk eligible, nothing pinned, yet the objective is lowest at the current assignment. This is a real, expected outcome, not a bug: it happens when the only move (or moves) that would meaningfully improve the spread costs more, in `moves`/`bytes`/`fragmentation`, than the imbalance it fixes is worth under the configured weights. Without this section, that case is indistinguishable from "the solver didn't try" — `plan`'s one-line verdict says `ACT` either way.

`explain` finds the single-disk move closest to being worth taking — every movable disk against every other storage in the group, evaluated with the same objective the solver itself minimizes (section 5.3's one-move neighbourhood) — and shows the term-by-term arithmetic that rejected it:

```
objective: imbalance 0.576 + moves 0 + bytes 0 + fragmentation 0 + spread 0 + split 0 + reserve 0 = 0.576
no moves made: the objective is lowest at the current assignment
closest alternative: vm110(110):scsi1 VM-krbd → VM
  imbalance 0.576→0.0426, moves 0→0.25, bytes 0→0.00732, fragmentation 0→0.5, spread 0→0, split 0→0,
  reserve 0→0
  total 0.576 → 0.8  (worse by 0.223 -- rejected)
```

Read left to right: moving `vm110(110):scsi1` from `VM-krbd` to `VM` would cut the imbalance term from `0.576` to `0.043` — a real improvement — but `110` is a two-disk VM with its other disk staying put, so the move splits it across two storages, and `objective.kappa_vm_affinity`'s fragmentation penalty (`0` → `0.5`) for that outweighs the gain. If this is a tradeoff you want to accept rather than reject, lowering `objective.kappa_vm_affinity` (or `objective.beta_move_count`/`gamma_move_bytes_per_tib`, whichever term dominates the rejection) is the knob — not a reason to suspect the solver.

Omitted entirely when there is nothing to compare against (every disk pinned, or a one-storage group) or when the plan actually moved something.

`measured load:`

The section 4 input every number above derives from — identical to `show-load`'s own per-storage, per-disk report (`docs/manual/25-show-load-and-verify-storages.md`): each storage's used/capacity, `L_s`/`u_s`, reserve status, and every disk on it with its size, format, measured `ℓ_d` when one was fetched, and `[pinned: ...]` when section 5.3 (C2) excludes it. Always present, printed even for a `NO ACTION` group and even when no migration was possible at all — it is the data the rest of `explain`'s narrative is *about*, not part of the plan itself.

`-v` the `data source:` line

One extra line, printed once for the whole run rather than once per group (every group here was computed against the identical selector and window): the exact section 3.4 node-scoping filter this run's queries carried (`(no node-scoping filter)` if none applied), and the `window.lookback`/`quantile`/`metrics.rate_window`/`metrics.step` settings the load above was computed from — see `docs/manual/10-configuration.md` for what each controls and `IMPLEMENTATION_PLAN.md` section 3.4 for the filter itself. This is the one piece of `explain`'s output that is about *how* the data was fetched rather than what it is, which is why it is the one thing here gated behind `-v` instead of always shown. By default this is the node-list alternation from this cluster's own node names; with `metrics.extra_selector` set, it is that instead:

```
$ pve-storage-drs -c /etc/pve/drs.yaml -v explain
data source: {nodename=~"pve01|pve02|pve03"}  window 1.0d lookback, quantile 0.95, rate_window 5.0m, step
5.0m

Group fc-tier1 → ACT: imbalance 255% exceeds gates.imbalance_threshold (20%)
  ...
```

(With `metrics.extra_selector: 'cluster="mycluster"'`, the same line instead reads `data source: {cluster="mycluster"}`.)

`pinned (not movable this run):`

Every disk section 5.3 (C2) excludes from this run's solve, in the same `name(vmid):device` order `show-load` lists disks, with its size, current storage, measured load (`ℓ`) and `ℓ/z` ratio when one is available, and its exact reason — a real snapshot or an unreferenced companion volume (section 3.7), an unapplied pending config change (section 3.8), a config exclusion, a still-running `gates.cooldown_per_disk`, a VM config lock, or a small disk (below `migration.tiny_disk_bytes`) whose VM has no larger disk in this group to move with. Identical text to what `show-load` already prints per disk as `[pinned: ...]`; gathered here into one block instead of interleaved with movable disks, because pins are `explain`'s subject, not an aside.

A pin with something to actually act on or wait for gets a trailing `→ <hint>` — "clear snapshots to unblock" for a real snapshot, "check that volume and, if it is a leftover copy, delete it by hand to unblock" for an unreferenced volume (the reason names it, as `storage:volume`), "apply the pending change (reboot the VM) or revert it, then re-check" for a pending config change, "re-check next run" for a cooldown, "re-check next run once the lock releases" for a VM lock. A standing policy exclusion (`exclude.*`, or a deliberately skipped `unusedN` disk) carries no hint — that pin is not something to unblock, it is a choice already made in the config.

`cannot fully consolidate:`

Names a VM whose disks — after everything this run's plan actually managed to schedule — still sit on more than one storage, *because* at least one of them is pinned. A VM legitimately spread across storages

by an ordinary, unpinned plan is not reported here; that is the plan working as intended, not a blocker. This is section 3.6's own example almost verbatim: "evacuating a storage completely is … not achievable [when] it is because of a section 3.7 snapshot or an explicit exclusion, never because of a device type" — `explain` names which one, and on which device, rather than a plan quietly leaving a stray volume behind.

`pinned load ... ; best achievable spread given pins: ...`

`report.warn_pinned_load_fraction` (default `0.25`) compared against this group's pinned load as a fraction of its total measured load. Below the threshold, as in the example above, this line is purely informative. Above it, a second, ⚠-prefixed line follows: the residual imbalance may be structural — enough load is pinned that no achievable placement of the *movable* disks alone gets the group much closer to balanced — rather than a sign the solver or the weights need attention. `best achievable spread given pins` is exactly the `after:` spread line above: every solve (heuristic or MILP) already fixes every pinned disk at its current storage and optimizes only the movable ones (section 5.1's `x_{d,sigma_0(d)} = 1` for `d` outside `D^mov`), so what the plan actually reached *is* the best that pinned set permits — clearing the pins named above is the only way to do better, not a bigger `objective` weight or a longer `solver.time_limit_seconds`. Omitted entirely for a group with no measured load at all (an idle group, or one `explain` could not get a load for).

A `NO ACTION` group still gets the measured-load/pinned/fragmentation/ pinned-load sections above — they describe the state of the group's disks, not the plan, so there is no reason to withhold them just because the gate found nothing to balance this run.

`--json` emits everything `plan --json` does (`docs/manual/27-plan.md`'s own field list) plus `objective` (the seven terms above, `null` when the gate said `NO ACTION`), `rejected_alternative` (`null` unless the "`no moves made`" case above applies, otherwise `disk_key`, `vmid`, `vm_name`, `device`, `from_storage`, `to_storage`, `baseline` and `objective` — each the same seven-term breakdown `objective` above serializes, for the current assignment and the candidate respectively — and `worse_by`, the difference between the two totals), `storages`/`disks` (the measured-load section above, identical shape to `show-load --json`'s own fields of the same name), `pinned_disks` (`disk_key`, `vmid`, `vm_name`, `device`, `current_storage`, `size_bytes`, `load`, `load_per_tib`, `reason`, `action_hint` — `null` for a standing policy exclusion, same rule as the human report's → line), `fragmentation` (a list of `{vmid, vm_name, blockers}`, each blocker an object with `device`, `disk_key`, `reason`), and `pinned_load` (`null` for an idle group, otherwise `pinned_load`, `total_load`, `fraction` and `warn_fraction`). Unlike the human report, JSON has no notion of `-v`: a top-level `query` object (`node_selector`, `window_lookback_seconds`, `quantile`, `rate_window_seconds`, `step_seconds`) is always present, once per response rather than once per group, the JSON counterpart of the `-v data source:` line above.

Safety properties, exit codes, and what this build actually does

What is safe, unconditionally

These hold regardless of `execution.mode`, and are not configurable away (`.agents/domain-invariants.md`):

- **Dry-run is the default.** A missing or unparseable `execution.mode` means dry-run, never automatic execution.
- **The snapshot reserve is never traded for balance, and neither is the free space you configure.** `pve-storage-drs` will leave a group imbalanced rather than breach `snapshot_reserve.factor`/`free_space.soft` on any storage, and that reserve holds *during* a migration, not merely before and after it — down to `free_space.hard`, never below it.
- **Nothing is ever deleted automatically.** An orphaned volume left by a failed move is reported, never removed. The operator deletes.

- **An explicitly named configuration that cannot be read or fails validation is a hard failure**, never a silent fallback to `/etc/pve/drs.yaml`.

Exit codes

Code	Meaning
`0`	Success. A run that stopped at a gate, found nothing worth moving, or (for `verify-metrics`) found no error-level finding.
`1`	The run failed: invalid configuration, an unreachable Prometheus or PVE API (including one that fails mid-run, for any group, in the first plan or in a re-plan), a failed move, a failed check, or a command not yet implemented in this build (see below). A run that *bails out* because `execution.max_replans_per_run` was exhausted is not a failure and exits `0`.
`2`	Usage error on the command line.

What this build actually implements

`IMPLEMENTATION_PLAN.md` section 12 lays out thirteen phases, all thirteen now implemented, including phase 10 (`collect-testdata`, `--replay` and the real-cluster test corpus of section 16), phase 11 (the section 2.3 logging policy — see `35-logging.md`; the status file a monitoring check reads is `36-monitoring.md`), phase 12 (the capacity-spread objective and gate, section 5.3 (C7)/5.4's δ term) and phase 13 (per-storage free-space requirements, section 5.3.1). Dogfooding against a production cluster continues, and `REVIEW.md` tracks findings from that to resolution — being honest about exactly where a still-evolving tool stands matters more than a document that reads as finished and closed:

Command	Status
`--version`, `--help`, `--manual`	Implemented.
`verify-metrics`	Implemented: all six `IMPLEMENTATION_PLAN.md` section 3.3 checks, human and `--json` output.
`show-load`	Implemented: every storage in every group, its disks (all buses), sizes, reserve status ((C4)/(C5)), per-disk/per-storage I/O load (`ℓ_d`/`L_s`/`u_s`, section 4), and a section 6 act/no-act verdict per group with its reasoning, pinned and low-coverage disks flagged with their reason (including a disk within `gates.cooldown_per_disk`, once `state.json` has recorded one), human and `--json` output. A Prometheus outage degrades this one group's load (and its gate verdict) to "unavailable" rather than failing the whole command — sizes and reserve status are unaffected. The gate verdict now reads real drift history from `state.json` when a group has one recorded; a group with none yet still evaluates as a first run (reserve override or imbalance only, never drift-suppressed) — see `docs/internals/15-state.md` and `docs/internals/80-gates.md`.
`verify-storages`	Implemented: `saferemove` and the implied wipe time per storage, the resolved `free_space.soft`/`.hard` bytes per storage (section 5.3.1), warning when `gates.cooldown_per_storage` or `migration.max_single_move_duration` is shorter than the implied wipe time.

Command	Status
`plan`	Implemented: per group, evaluates the section 6 gate (reading real drift history from `state.json` when a group has one recorded — same as `show-load`, see `docs/internals/15-state.md`), and on `ACT` solves it with `solver.backend` (`auto` tries CBC, then the dependency-free heuristic — see `docs/internals/91-optimize.md`; the MILP backend and the heuristic exclude, as a migration target, any storage within `gates.cooldown_per_storage`, except when repairing a live reserve violation, which is never deferred for a cooldown, in the heuristic only — see `docs/internals/91-optimize.md`'s own note on that gap), orders its moves under the section 8 transient reserve invariant, and checks the whole plan against section 7's payback (cost/benefit) test — a plan whose executed endpoint reduces the group's reserve/free-space shortfall (section 5.3.1's `free_space.soft`) is exempt from the economic half of that test but not from the hard per-move duration rule. Human and `--json` output report which backend actually solved each group. **No automatic re-solve-and-shrink on a failing payback test** — a plan that fails is reported, not silently adjusted (section 7.3's 3-retry loop is not implemented). See `docs/manual/27-plan.md` and `docs/internals/90-heuristic.md/91-optimize.md/95-schedule.md/96-payback.md`.
`explain`	Implemented: runs the identical per-group gate/solve/schedule/payback pipeline `plan` does (`_plan_group()`), then narrates what `plan`'s own output does not print. Lists every disk pinned this run with its exact reason (section 3.6/3.7 — snapshots, an exclusion, a lock, or a still-running `gates.cooldown_per_disk`), names any VM a pin leaves with disks spread across more than one storage ("cannot fully consolidate"), and breaks the section 5.4 objective down into its seven terms (`ObjectiveBreakdown`'s own reason for keeping them separate rather than only a `total`) when the gate acted. Also reports pinned load as a fraction of the group's total against `report.warn_pinned_load_fraction`, flagging when the residual imbalance is likely structural (too much load pinned to fix by moving anything) rather than a planning shortfall — alongside the best achievable spread the plan actually reached given those pins. Human and `--json` output, same shape as `plan`'s with these fields added. See `docs/internals/40-cli-and-logging.md` and `docs/internals/90-heuristic.md`.

Command	Status
`apply`	Implemented for all three `execution.mode` values. Before planning anything, checks for a `move_disk` left running by a crashed previous run or a second concurrent instance (section 13) and excludes its vmid from this run; runs the identical per-group gate/solve/schedule/payback pipeline `plan` prints, then executes it: pre-flight re-checks each move against the live cluster immediately before issuing it (including exclusion tags/vmids), waits out a VM config lock rather than failing, and does not consider a move done until the task succeeds *and* the source volume is gone *and* the VM's lock is clear — a `saferemove` wipe can hold those apart for hours (`draining`, excluded as both source and target for the rest of that run). `state.json`'s `inflight_upids` is written before each `move_disk` and cleared once its task finishes, so a crash mid-move leaves a trace the next run's own startup check finds. `auto` additionally honours `execution.time_windows` (local host time; refuses a move that cannot finish before the window closes) and `execution.max_migrations_per_run` (a shared budget across every group the run visits), and re-invokes the whole pipeline from freshly observed state on a `replan_needed` mismatch, bounded by `execution.max_replans_per_run` (section 9.2's re-plan protocol). `execution.max_concurrent_migrations`/`max_concurrent_per_storage` above their default of `1` are honoured in `auto` mode: section 8.1's generalized transient invariant and section 9.2's "poll all in-flight UPIDs" loop, strictly FIFO (never reordering the scheduler's own queue to keep every slot busy) — `dry-run`/`confirm` always run strictly sequentially regardless. See `docs/manual/28-apply.md`, `docs/internals/92-execute.md` and `docs/internals/93-crashrecovery.md`.
`collect-testdata`, and the global `--replay`	Implemented (`IMPLEMENTATION_PLAN.md` section 16, phase 10). `collect-testdata` captures everything the engine can read from the PVE API and Prometheus — the superset of every forecaster's data range, not just the configured one — replaces every identifier with a pseudonym keyed to a salt that never leaves the host (`support.salt_path`), and writes a bundle (directory + deterministic `.tar.gz`) you can read before deciding whether to send it to the author as test data; `--estimate` prints the query/sample-point estimate and refuses before fetching above `support.max_series_points`. `--replay BUNDLE` then runs `verify-metrics`, `verify-storages`, `show-load`, `plan` and `explain` against such a bundle with no network access at all, using the bundle's own `config.yaml` unless `-c` overrides it; `apply` and any `--mode` above `dry-run` are a usage error under `--replay` (exit 2), and so is `collect-testdata` itself, which needs a live cluster. See `docs/manual/26-collect-testdata-and-replay.md` and `tests/corpus/README.md`, which documents what a bundle does and does not hide and the terms on which one is accepted into the project.

Configuration loading and validation (this whole manual's `10-configuration.md`) is complete and exercised by every command: a config error is reported and the run exits `1` before `main()` ever dispatches to a command's own handler, so validating a configuration file does not require the rest of the engine to be involved at all.

Every command in the table above now has a real handler, but the dispatch mechanism that reports one honestly if it did not would still print a message naming the plan section it belongs to, and exit `1` — `docs/internals/40-cli-and-logging.md` describes it as the extension point for the next new command this codebase adds, not dead code kept around for its own sake.

Optional dependencies

`pve-storage-drs` runs, plans and executes — every `execution.mode` (`dry-run`/`confirm`/`auto`) included — with only `requests`, `ruamel.yaml` and `jsonschema` installed. One dependency is still optional and is imported only where it is used, never at module level:

- **`statsmodels`** (Debian `Suggests`) — needed only for `forecast.model: holt_winters`. Without it, that model logs a warning and falls back to `quantile` automatically rather than failing.

`pulp` + **`coinor-cbc`** are a Debian `Depends`, not a `Recommends`: on a Debian install a real MILP solver — CBC through `pulp` — is always present, and `solver.backend: auto` (the default) uses it. The dependency-free heuristic of `IMPLEMENTATION_PLAN.md` section 5.5 is still a genuine fallback, not dead code — it runs when a non-Debian install has no solver extra, when a solve fails or times out, or when `solver.backend: heuristic` is configured explicitly — but it is no longer what a plain `apt install pve-storage-drs` lands on by default. (An optional `ortools`-based CP-SAT backend existed through release 0.1.8; it was removed — `ortools` has no Debian package, so that backend could never run on a PVE host. See `docs/internals/91-optimize.md`.)

Logging: what lands where, and what an unattended run records

Two rules cover almost everything:

- **The report goes to stdout, the log goes to stderr.** Always, in every mode. `pve-storage-drs --json plan > plan.json` is safe: nothing is ever interleaved into that file.
- **A run in which nothing went wrong logs nothing.** By default only warnings and errors are logged, so a healthy `plan`, `explain` or `show-load` prints its report and not one line beside it.

Verbosity

	Level	What you get
`--quiet`	errors	The run failed, nothing else
(default)	warnings	Something degraded but the run continued
`-v`	info	This run's decision trail (below)
`-vv`	debug	Per-query detail, plus `urllib3`/`proxmoxer` library logs

`--log-level error|warning|info|debug` states a level outright and wins over both `-v` and `--quiet` — for automation that would rather name a level than count vs. The one exception is the mandatory floor below ("Unattended runs log this without being asked"): `--log-level warning`/`error` on a `confirm`/`auto` apply run is raised back up to `info` rather than silently discarding the audit trail. `--quiet` is the only flag that discards it.

The decision trail

`-v` logs, per group, the reasoning behind what the report shows: the measured load (`load_digest`), the gate's computed drift/imbalance together with the thresholds they were compared against (`gate_decision`), the chosen plan with its solver backend and the six objective terms (`plan_selected`), section 7's payback arithmetic and the ratio it needed (`payback_verdict`), and every migration issued with its PVE

task UPID (`move_started` / `move_finished`). A run opens with `run_started` and closes with `run_summary` — moves issued, succeeded, failed, bytes moved, wall time and exit code.

```
$ pve-storage-drs -v plan
run started: plan (dry-run)
group fc-tier1: load 7.400 across 6 disks
group fc-tier1: ACT -- imbalance 255.0% meets or exceeds gates.imbalance_threshold (20.0%)
group fc-tier1: cbc plan, 2 move(s)
group fc-tier1: payback ratio 151 (need 10) -> accepted
run finished: plan exit=0
```

Unattended runs log this without being asked

`apply` in `confirm` or `auto` mode emits the whole decision trail above **whether or not `-v` was given**. In those modes the log is the only record of what happened to your cluster, and a record you only get if you remembered a flag is not a record. Read-only commands and `apply --mode dry-run` change nothing, so they stay at the ordinary default.

`--quiet` still wins, including over this. It is the one option that will leave you with an `auto` run that migrated data and no account of what it moved or why — which is why a timer unit should not pass it.

Text or JSON

`--log-format text` (the default) writes **one human-readable line per record**, at a terminal and in the journal alike, so `journalctl -u pve-storage-drs` reads like the terminal output:

```
$ pve-storage-drs -v plan 2>&1 >/dev/null | head -1
run started: plan (dry-run)
```

`--log-format json` writes **one JSON object per line** instead, for a pipeline that filters on it:

```
$ pve-storage-drs --log-format json -v plan 2>log.json >/dev/null ; head -c 120 log.json
{"command": "plan", "event": "run_started", "level": "INFO", "logger": ...
```

Every JSON record carries an `event`, which is the supported way to filter:

```
journalctl -u pve-storage-drs.service -o cat | jq -r 'select(.event=="move_started") | .upid'
```

This is the one stream written *as each move starts and finishes*, UPID included; the `--json` report on stdout only appears once the run is over, so a run that dies mid-move never prints it. That is the reason to choose JSON for an unattended timer.

`--log-format auto` was the default through 0.1.9 and chose JSON whenever stderr was not a terminal, which included the journal. It is still accepted and now means `text`; a unit that relied on it for JSON needs `--log-format json`.

Under systemd

Point `StandardError=` at the journal and let the defaults do the rest: `auto` mode raises its own verbosity to log the audit trail, as text. Add `--log-format json` if something downstream parses the journal. Do not add `--quiet`; do not add `-v` (it is only needed if you also want the trail from a `dry-run` timer).

Monitoring: the status file

A timer-driven `pve-storage-drs apply --mode auto` that fails quietly is worse than one that fails loudly, and the journal is not a monitoring system. Set `monitoring.status_file` and `apply` leaves a small report of its last run in the format of the **`check_statusfile`** plugin from Debian's `monitoring-plugins-contrib`

package (`/usr/lib/nagios/plugins/check_statusfile`). Nagios, Icinga, or anything else that runs a Nagios plugin can then watch it with nothing of ours installed on the monitoring side:

```
/usr/lib/nagios/plugins/check_statusfile /var/lib/pve-storage-drs/status
/usr/lib/nagios/plugins/check_statusfile -a 1h /var/lib/pve-storage-drs/status
```

`-a` is the maximum age of the file (`s`, `m`, `h` or `d`; default `26h`) — see "Freshness" below.

What is in the file

A worked example of a run that gave up after too many outside changes:

```
WARNING
apply completed with 2 warnings: group fc-tier1: execution.max_replans_per_run (3) exceeded (last mismatch:
VM 101 is now on 'san-c', not the planned 'san-a') | groups=1 moves_succeeded=2 moves_failed=0
bytes_moved=2199023255552B replans=3 warnings=2 duration=412.6s
warning: group fc-tier1: 9.50 GiB short of its snapshot reserve / free-space requirement after this run's
plan
pve-storage-drs 0.1.13, mode auto, finished 2026-09-20T20:30:00Z
```

- **Line 1** is the state: exactly `OK`, `WARNING` or `CRITICAL`. The plugin turns it into its exit status (`0`, `1`, `2`).
- **Line 2** is the summary, naming the first problem if there is one. After the `|` come `label=value` performance data your monitoring can graph: `groups`, `moves_succeeded`, `moves_failed`, `bytes_moved` (bytes), `replans`, `warnings`, `duration` (seconds).
- **The rest** lists any further errors and warnings, one per line (at most 20, then "and N more — see the run log"; a message longer than 300 characters is cut with `...`, since monitoring systems truncate plugin output — the full text is in the run log), and ends with the version, the mode and the finish time. The file holds counts, group and storage names and volume ids — never credentials.

What makes a run `OK`, `WARNING` or `CRITICAL`

State	When
`CRITICAL`	The run exited non-zero: a move failed; PVE or Prometheus could not be read (for any group, in the first plan or in a re-plan — the run stops at once); or the tool crashed (the file says `CRITICAL` before the crash propagates, so a bug cannot leave the previous run's `OK` in place).
`WARNING`	The run exited `0` but left something for a human: it gave up after `execution.max_replans_per_run` re-plans because outside changes kept invalidating its plans (try again later); a failed move left a volume on the target, which is reported and **never deleted**; a source storage had not released a volume after a move (`draining`, usually a slow `saferemove` wipe); or a group is **still short of its snapshot reserve or free-space requirement** once this run's plan has run — the tool never trades the reserve for balance, so it will not fix that by itself.
`OK`	None of the above. A run that found nothing worth moving is `OK`; so is a `dry-run`.

`UNKNOWN` is never written by the tool. You will see it from the plugin itself when the file does not exist yet, or is not in the expected shape.

Freshness: why every run rewrites the file

`check_statusfile` judges freshness by the file's **modification time**, not its content: older than `-a` is reported `WARNING`. So `apply` rewrites the file on **every** run that produced a result — including runs that moved nothing and dry runs — so that a stale file means "the timer stopped", never "it was a quiet day".

Two cases deliberately leave the file alone: a run that exited quietly because another instance held the lock (it produced no result, and the file keeps the last real one), and a run that could not read its configuration at all (there is no path to write to; the file simply goes stale). `plan`, `explain`, `show-load` and `--replay` never write it: an operator running them by hand must not overwrite the timer's last result.

The file is written to a temporary file in the same directory and renamed into place, so the plugin can run at any moment without reading half a report. If it cannot be written at all, the run carries on and exits as it would have (a status file that cannot be written must not change what a run did), a `status_file_write_failed` error is logged, and the file turns `WARNING` by age on its own.

Wiring it into Nagios/NRPE

The check runs on the node that owns `state.json` — the one node the timer runs on:

```
command[check_pve_storage_drs]=/usr/lib/nagios/plugins/check_statusfile -a 1h
/var/lib/pve-storage-drs/status
```

Pick `-a` a little longer than the timer's period. Too generous and a stopped timer goes unnoticed for that long; too tight and one slow run turns the check `WARNING`.